BA ZC581 :
Applied Data Science
Module: 13

BITS Pilani

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J A Gokhale.

Disclaimer

- The slides presented here are obtained / extracted from the authors of the prescribed books and from various other subject contributors.
- I hereby thank and acknowledge all the contributors for their material and inputs.
- I have added and modified a few slides to suit the requirements of the course.

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Module 13:
Business Driven KPIs and Metrics

Module 13 Topics

Topic 1: Designing and Analysing KPIs using Data

Topic 2: Example KPIs from a few domains

Topic 3: ROI, Retention, Churn, Throughput, Absenteeism

Topic 4: How to present KPIs

Ref: T1 , Ch 15

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Topic 1: Designing and Analysing KPIs using Data

Performance Indicators (PIs)

- Performance Indicators are **general measurements** that track activity, behavior, or performance.
  - Also called **metrics**
- They are useful, but **not always tied to a strategic goal**
- Examples:
- Website Bounce Rate:
  - Measures the percentage of visitors who leave after viewing only one page. Helps understand engagement quality.
- Average Resolution Time (Support Desk):
  - Measures how long it takes, on average, to resolve a customer ticket. Indicates operational efficiency

Performance indicators (PIs)

Also known as Metrics, are quantifiable measures used to assess the status of the business, realized after computing on measured values. Metrics can be directly aggregated, computed, or otherwise deduced.

Check these

- Average sale amount
- Maximum hours worked
- Total units produced
- Customers acquired per 1000 mails sent
- Count of Employees
- Number of times temperature crossed limit

And these

- Names of vendors delisted
- List of active suppliers
- Top 3 performers by sales
- Department names
- Order Status Shipped / Not Shipped

Performance Indicators

- Non-PIs, but used for analysis
  - Top Selling Product of the Month – Descriptive PI, not linked to business goal
  - Top 3 Product by sales – Descriptive PI, not linked to strategic target
  - Names of Vendors Delisted – Operational Metric, not connected to any business goal or strategic target

If it **doesn't answer "how much?", "how many?", "how fast?", "how often?", or "what %?"**, then it is not a metric.

Key Performance Indicators aka KPIs

- KPIs are **strategic, goal-linked metrics** that directly show whether the business is achieving its objectives.
- All KPIs are metrics, but not all metrics are KPIs
- Usually expressed as
  - A percentage change over previous observation
  - A relative trend direction
  - A ratio parameter / factor
  - Against a set target

Check these

- Increase in sales of Shampoos by 15% this quarter against a target of 10%.
- Monthly growth in number of new customers added %
- Number of calls converted in current quarter is 1.35x of previous quarter

And these

- Average sales invoice value
- Number of new customers added
- Daily average of calls converted per executive = 26

Example

Consider the data below:

Product	Units Sold	Revenue
A	120	24000
B	80	16000
C	150	30000
D	50	10000

Metric: Total Revenue
We simply calculate what happened — no target, no goal.
Total Revenue=24,000+16,000+30,000+10,000=80,000
Interpretation: The business generated ₹80,000 in revenue in January.

KPI: Achieve ₹1,00,000 revenue in January
Now we compare actual performance to a target.
KPI Achievement % = (80,000/1,00,000) * 100=80%
Interpretation: The business achieved 80% of its revenue target.

Example, contd

- **Metric: Top-selling product by revenue**

  Product C = ₹30,000 → highest

  This is descriptive → metric

- **Target: Increase sales of Product C by 10% next month**

  Target = 30,000 * 1.10=33,000

  If next month revenue = ₹31,000:

  KPI Achievement% =(31,000/33,000)*100 =93.9%

KPI Reporting

- KPI Reporting should to be actionable, meaningful, understandable, and accurate.
- Each business function and department will have different (and unique) key performance indicators.
- Develop unique KPIs representative of the specific business function

Example PIs & KPIs

- Work Order Analysis
  - Work Order Age Report
  - No of Open Work Orders
  - Work Order Cycle Times
  - Late work order %
  - Backlog Quantity
  - New Work orders added this month
  - Completed Work Orders %
  - Production Attainment Monthly
  - 0-30 days Demand Quantity
  - 0-30 days Supply quantity

Important Aspects of Selecting KPIs

1. It aligns with the company's goals

   All organizations have objectives they are striving towards. KPIs should align with the company's goals to help measure success towards those goals.

2. It is specific and measurable

   With any KPI, it must define and measure something tangible so one can see if we are on track towards the goals. KPIs should be as specific as possible to be useful.

3. It provides achievable value

   A KPI must be something that is achievable to provide true value. It has to be actionable, meaning the data it provides is used to make improvements as needed.

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Example KPIs from various segments of the organization

Example: Manufacturing KPIs

1. On-time Delivery
2. Production Schedule Attainment
3. Total Cycle Time
4. Throughput
5. Capacity Utilization
6. Changeover time
7. Yield
8. Scrap
9. Planned Maintenance
10. Availability
11. Customer Return Rate (Rejection Rate)
12. Overall Equipment Effectiveness

Example

Courtesy: Proaction International

Check the link

https://insightsoftware.com/blog/30-manufacturing-kpis-and-metric-examples/

Finance KPIs

1. Profitability KPIs

Gross Profit Margin
Measures profit after direct costs.
Formula: (Revenue − COGS) / Revenue

Operating Profit Margin (EBIT Margin)
Shows core business profitability.

Net Profit Margin
Final profitability after all expenses.

EBITDA Margin
Profitability before interest, tax, depreciation, amortization.

ROA (Return on Assets)
How efficiently assets generate profit.

ROE (Return on Equity)
Profit generated per unit of shareholder equity.

ROCE (Return on Capital Employed)
Measures returns from long-term capital.

Other Finance KPIs

2. Other KPIs

Operating Cash Flow Ratio
Cash flow strength vs liabilities.

Working Capital Turnover
Revenue generated per unit of working capital.

Asset Turnover Ratio
Efficiency of asset usage.

Inventory Turnover
How often inventory is sold and replaced.

Operating Cash Flow
Cash generated from core operations.

ARPU (Average Revenue per User)
Revenue efficiency per customer.

Customer Lifetime Value (CLV)
Total value expected from a customer.

Sales Forecast Accuracy
Predictability of revenue.

COGS % of Revenue
Direct cost efficiency.

Operating Expense Ratio
Overheads as a share of revenue.

Debt-to-Equity Ratio
Leverage and financial risk.

Sales KPIs

Monthly Sales Revenue Target Achievement
Sales Growth Rate
Lead to Opportunity Conversion Rate
Opportunity to close Rate
Average Deal Size
Sales Cycle length
Customer Acquisition Cost (CAC)
Customer Lifetime Value (CLV)
Churn Rate
Customer retention rate
Sales pipeline value
Win rate
Sales per representative
Sales target achievement %

HR KPIs

Cost per Hire
Offer Acceptance Rate
Workforce Utilization Rate
Revenue per employee
Attrition rate
Early attrition rate (first 6-12 month)
Average Employee Tenure
Benefits Costs / employee
Overtime Rate / Overtime hours per employee
Salary variance to market
Absenteeism Rate
Leave Utilization rate
Training Effectiveness Score
Skill gap Index
Training cost per employee
Training completion rate
Disciplinary Case rate
Workplace incident Rate
Gender Diversity Rate

Purchase KPIs

Purchase cost Variance (PCV)
YoY Cost Reduction
Cost per PO
Price Variance
Supplier Lead Time
Supplier Defect Rate / Quality Score
Requisition to Order Cycle Time
PO Processing time
Inventory Turnover
Material Availability Rate
Procurement as % of Revenue
Procurement cost as % of spend
Early Payment Discount Capture Rate
Local Sourcing %
Single Source Dependency %

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Topic 2: Example KPIs from a few Domains

Domain specific KPIs

- Each domain of activity or industry has different goals; hence need different KPIs
  - Manufacturing KPI could be: reduce scrap ratio to < 1%
- Domain-specific KPIs are goal-linked performance indicators designed for a particular function (like Sales, HR, Finance) or industry (like Manufacturing, Retail, Healthcare).
  - Health care sales would be a variant from Consumer goods sales and hence different KPIs
- They measure success in the context of that domain's priorities, processes, and outcome.
- Not generic but tailored

CRM KPIs

Customer Acquisition Cost (CAC)
Cost per Lead(CPL)
Lead Conversion Rate
Customer Retention Rate
Customer Lifetime Value (CLV)
Customer churn rate
Customer Satisfaction Score (CSAT)
Average Resolution Time
Escalation Rate
Revenue per Customer
Upsell / Cross-sell rate
Social Engagement Rate
Average Response Time
Ticket Backlog
Cost to Serve
Revenue growth from existing customers

Healthcare KPIs

Average length of Stay (ALOS)
Readmission rate (30-day)
Mortality rate
Patient Satisfaction score (HCAHPS)
Patient Complaint Rate
Bed Occupancy Rate
Bed turnover Rate
Operation Room Utilization Rate
Referral conversion rate
Revenue per patient
Cost Per patient / Operating Margin
Days in Accounts Receivable
Incident Reporting Rate
Data privacy compliance (HIPPA adherence)
Electronic Health Record (HER) adoption rate
Chronic Disease management Rate
Drug inventory turnover

E-Commerce KPIs

Website Traffic (Sessions/ Users)
Cost Per click
Customer Acquisition Cost (CAC)
New and Returning visitor's rate
Add to cart rate
Cart Abandonment Rate
Average Order Value (AOV)
Revenue Per visitor (RPV)
Units per Tx (UPT)
Customer Lifetime Value (CLV)
Churn Rate
Time between Purchases
Return on Ad Spend (ROAS)
Affiliate Conversion Rate
Cost of Goods Sold (COGS)
Average Discount %
Order Fulfilment time
On-time delivery rate
Return Rate
Page Load time
Product View to Purchase Rate

QC & Inspection KPI

Defect rate
PPM Defects
First Pass Yield
First Time Right
Rejection Rate
Final Inspection pass rate
Inspection coverage %
False rejection rate
Escape rate
Supplier defect rate
Supplier quality score
Cost of Poor Quality (COPR)
Internal / External failure cost
Rework cost as % of sales
Inspection Cycle time
Inspector productivity (units per day)
Root cause recurrence rate
Customer complaint rate
Customer Resolution time
Reliability rate

Individual (ex: salesperson) KPIs

Sales Revenue (M/Q/YTD) growth
% Target Achieved
New Revenue Generated
Average Deal Size
Pipeline value
Pipeline conversion Ratio
Opportunity to close rate (Win rate)
Lost deal %
Calls per day/week
Meetings / week
Demos / week
Proposal Submission Rate
Customer retention rate
Repeat business rate
Account penetration rate
Revenue per client
Travel cost to Revenue ratio
New account acquisition rate

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Topic 3: ROI, Retention, Churn etc

A Few Examples

- KPIs need to be computed.
- Computational Logic should be clear and accurate
- We shall work on a few
  - ROI
  - Retention
  - Churn
  - Throughput
  - Absenteeism

Return On Investment (ROI)

Identifies the profitability or efficiency of an investment. Gain or loss relative to an investment. Does not consider time period, risk etc

Return on Investment (ROI)

- **Basic Formula**

$$\mathbf{ROI} = \frac{\text{Net Profit}}{\text{Investment Cost}} \times 100$$

Where:

$$\text{Net Profit} = \text{Total Gain from Investment} - \text{Total Investment Cost}$$

Example:

A company invests **₹5,00,000** in a marketing campaign.

The campaign generates **₹8,00,000** in revenue.

Assume total campaign-related cost is ₹5,00,000.

Step 1: Calculate Net Profit

$$\text{Net Profit} = 8,00,000 - 5,00,000 = 3,00,000$$

Step 2: Calculate ROI

$$\text{ROI} = \frac{3,00,000}{5,00,000} \times 100 = 60\%$$

✅ **ROI = 60%**

This means the company earned 60% return on the amount invested.

Retention

Identifies how effectively the business keeps its existing customers / employees.

🔹 Customer Retention Rate (CRR) Formula

$$\textbf{Customer Retention Rate (\%)} = \frac{\text{Customers at End of Period} - \text{New Customers Acquired}}{\text{Customers at Start of Period}} \times 100$$

🔎 What Each Term Means

- **Customers at Start (S)** → Total customers at beginning of period
- **Customers at End (E)** → Total customers at end of period
- **New Customers (N)** → Customers acquired during the period

So it is often written as:

$$\mathbf{CRR} = \frac{E - N}{S} \times 100$$

Example:

A company starts the quarter with **500 customers**.

During the quarter, it acquires **120 new customers**.

At the end of the quarter, it has **540 customers**.

Step 1: Apply Formula

$$CRR = \frac{540 - 120}{500} \times 100$$

$$CRR = \frac{420}{500} \times 100$$

$$CRR = 84\%$$

📌 Interpretation

The company retained **84% of its existing customers** during the quarter.

Churn

Identifies percentage of customers who stop doing business with the company over a specific time (M/Q/YTD). Indicates loyalty.

- **Customer Churn Rate Formula**

$$\textbf{Churn Rate (\%)} = \frac{\text{Customers Lost During the Period}}{\text{Customers at the Start of the Period}} \times 100$$

Example:

A company starts the month with **1,000 customers**.

During the month, **80 customers cancel** their subscription.

$$\text{Churn Rate} = \frac{80}{1000} \times 100 = 8\%$$

✅ **Monthly Churn = 8%**

Throughput

Measures the total number of units produced / processed by the business within a specific time frame, revealing operational efficiency, capacity and bottlenecks.

✅ **Basic Throughput Formula**

$$\textbf{Throughput} = \frac{\text{Number of Units Produced}}{\text{Time Period}}$$

It measures how many units a system, machine, or process produces in a given time.

Example:

A production line produces **1,200 units** in **8 hours**.

$$\text{Throughput} = \frac{1200}{8}$$

$$\text{Throughput} = 150 \text{ units per hour}$$

✅ **Throughput = 150 units/hour**

Example:

A factory produces **4,500 units** in a 9-hour shift.

$$\text{Throughput} = \frac{4500}{9}$$

$$\text{Throughput} = 500 \text{ units per hour}$$

Absenteeism

A HR KPI that measures how often employees miss work, revealing patterns that signal low morale, burnout, or issues with health / wellness programs

$$\textbf{Absenteeism Rate (\%)} = \frac{\text{Total Absent Days}}{\text{Total Available Work Days}} \times 100$$

Where:

$$\text{Total Available Work Days} = \text{Number of Employees} \times \text{Working Days in Period}$$

Example:

A company has **50 employees**.

There are **22 working days** in the month.

Total recorded absent days during the month = **40 days**.

Step 1: Calculate Total Available Work Days

$$50 \times 22 = 1100 \text{ total available days}$$

Step 2: Apply Formula

$$\text{Absenteeism Rate} = \frac{40}{1100} \times 100$$

$$= 3.64\%$$

✅ **Monthly Absenteeism Rate = 3.64%**

Overall Equipment Effectiveness

Identifies the percentage of manufacturing time (or resources) that is truly used productively

$$\mathbf{OEE} = \text{Availability} \times \text{Performance} \times \text{Quality}$$

Where:

1. **Availability**

$$\text{Availability} = \frac{\text{Operating Time}}{\text{Planned Production Time}}$$

2. **Performance**

$$\text{Performance} = \frac{\text{Ideal Output}}{\text{Actual Operating Time}}$$

(or)

$$\frac{\text{Actual Output}}{\text{Target Output}}$$

3. **Quality**

$$\text{Quality} = \frac{\text{Good Units Produced}}{\text{Total Units Produced}}$$

Example:

A machine operates in an 8-hour shift (480 minutes).

- Planned Production Time = 480 minutes
- Downtime = 60 minutes
- Actual Output = 350 units
- Ideal Production Rate = 1 unit per minute
- Defective Units = 20

Overall Equipment Effectiveness (contd)

Step 1: Availability

$$\text{Operating Time} = 480 - 60 = 420 \text{ minutes}$$

$$\text{Availability} = \frac{420}{480} = 0.875$$

Step 2: Performance

Ideal output in 420 minutes:

$$420 \text{ units}$$

$$\text{Performance} = \frac{350}{420} = 0.833$$

Step 3: Quality

$$\text{Good Units} = 350 - 20 = 330$$

$$\text{Quality} = \frac{330}{350} = 0.943$$

Step 4: OEE Calculation

$$\text{OEE} = 0.875 \times 0.833 \times 0.943$$

$$\boxed{\text{OEE} = 0.687 = 68.7\%}$$

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Topic 4: How to Present KPIs

Start with the Context

- KPI should be telling a performance story clearly, visually, and actionably
  - What objective are we measuring?
  - What decision will this KPI support?
  - Who is the audience (CEO, functional head, team member)?
- Example:
  Instead of saying:

  "Our churn rate is 8%"

  Say:

  "Customer retention is weakening, impacting recurring revenue stability."

Use a clear KPI structure

A good KPI presentation follows this format:

- KPI Name
- Definition
- Formula
- Target
- Actual
- Variance
- Trend
- Action Plan

example

KPI	Target	Actual	Variance	Trend	Status
Revenue Growth	10%	8%	-2%	↓	⚠️
Churn Rate	5%	8%	+3%	↑	🔴
OEE	80%	82%	+2%	↑	🟢

Show Trends, not just numbers

- Never present a single-period number.
- Always show as a percentage variance:
  - Month-on-month
  - Quarter-on-quarter
  - Year-on-year
- Line graphs work best for trend KPIs like:
  - Revenue
  - Absenteeism
  - Churn
  - Production output

Use Right Visualization type

Create a dashboard to showcase the KPIs as visuals

KPI Type	Best Visual
Target vs Actual	Gauge / Bullet Chart
Trend	Line Chart
Comparison	Bar Chart
Composition	Stacked Bar / Pie
Efficiency	Ratio Card
Operational Health	Traffic Light Indicators

Highlight Variance & Root casue

Managers care about:

- What changed?
- Why did it change?
- What are we doing about it?

Structure:

Observation → Cause → Action

Example:

> OEE dropped from 78% to 68%.
> Root cause: Increase in downtime due to maintenance delays.
> Action: Preventive maintenance schedule revised.

Align KPIs to strategy

- KPIs must link to business goals:
  - Revenue KPIs → Growth Strategy
  - Quality KPIs → Customer Strategy
  - Cost KPIs → Efficiency Strategy
  - Employee KPIs → Capability Strategy
- If a KPI is not linked to strategy, remove it

Keep it Simple

For leadership:

- 5–10 KPIs maximum
- One page dashboard
- Red-Amber-Green indicators
- Clear next steps

Avoid:

- Too many metrics
- Technical jargon
- Overcrowded charts

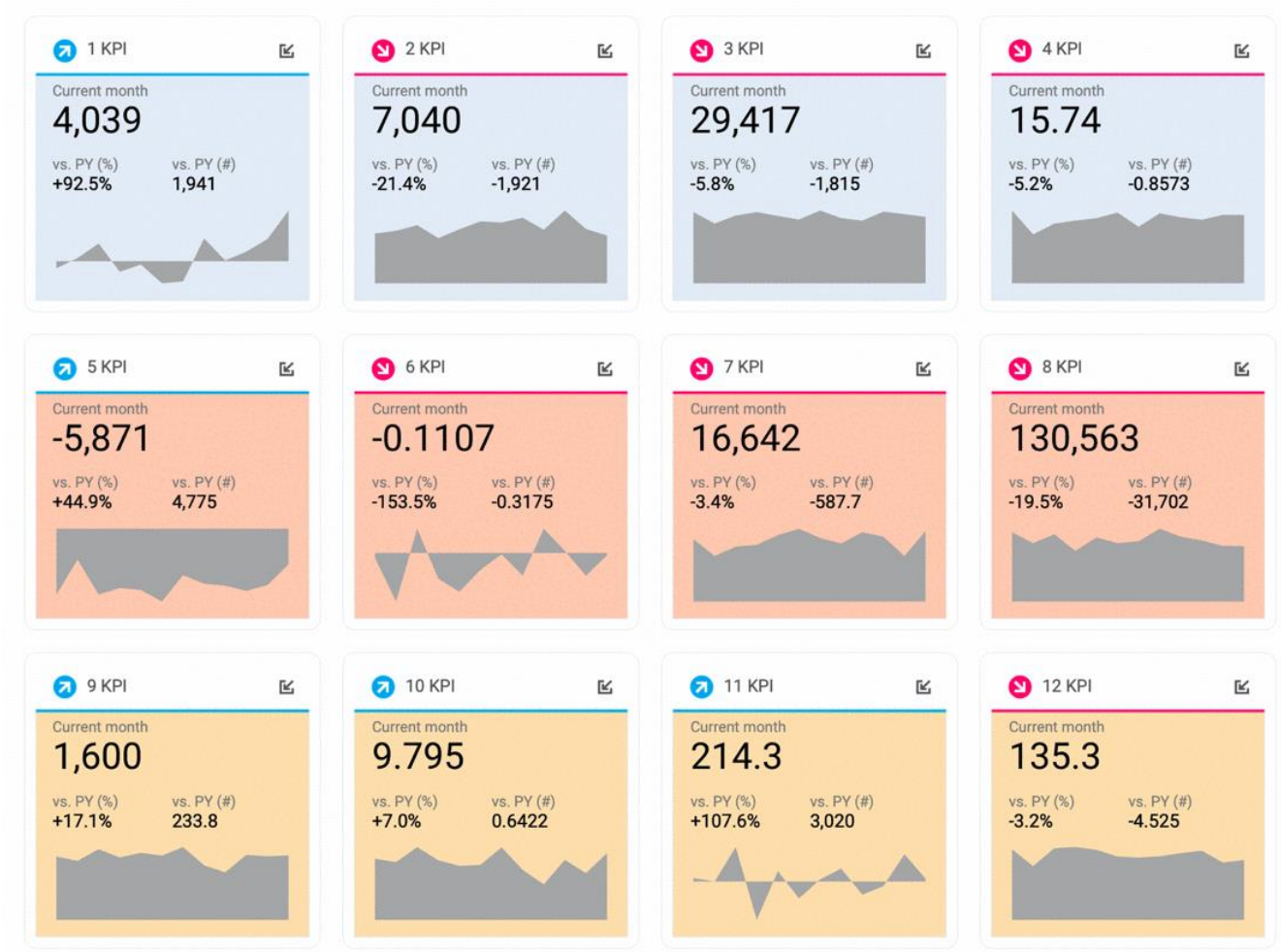
Dashboard Layering Approach

- **Level 1: CEO Dashboard**
  - 5–8 Strategic KPIs
  - Filters / Slicers
  - 6-8 Visualizations
- **Level 2: Functional Dashboard**
  - Department KPIs
  - Filters / slicers
  - 8-10 visualizations
- **Level 3: Operational Dashboard (sometimes also called a KPI dashboard)**
  - 8-10 Daily process KPIs
  - Progress funnels
  - Backlog

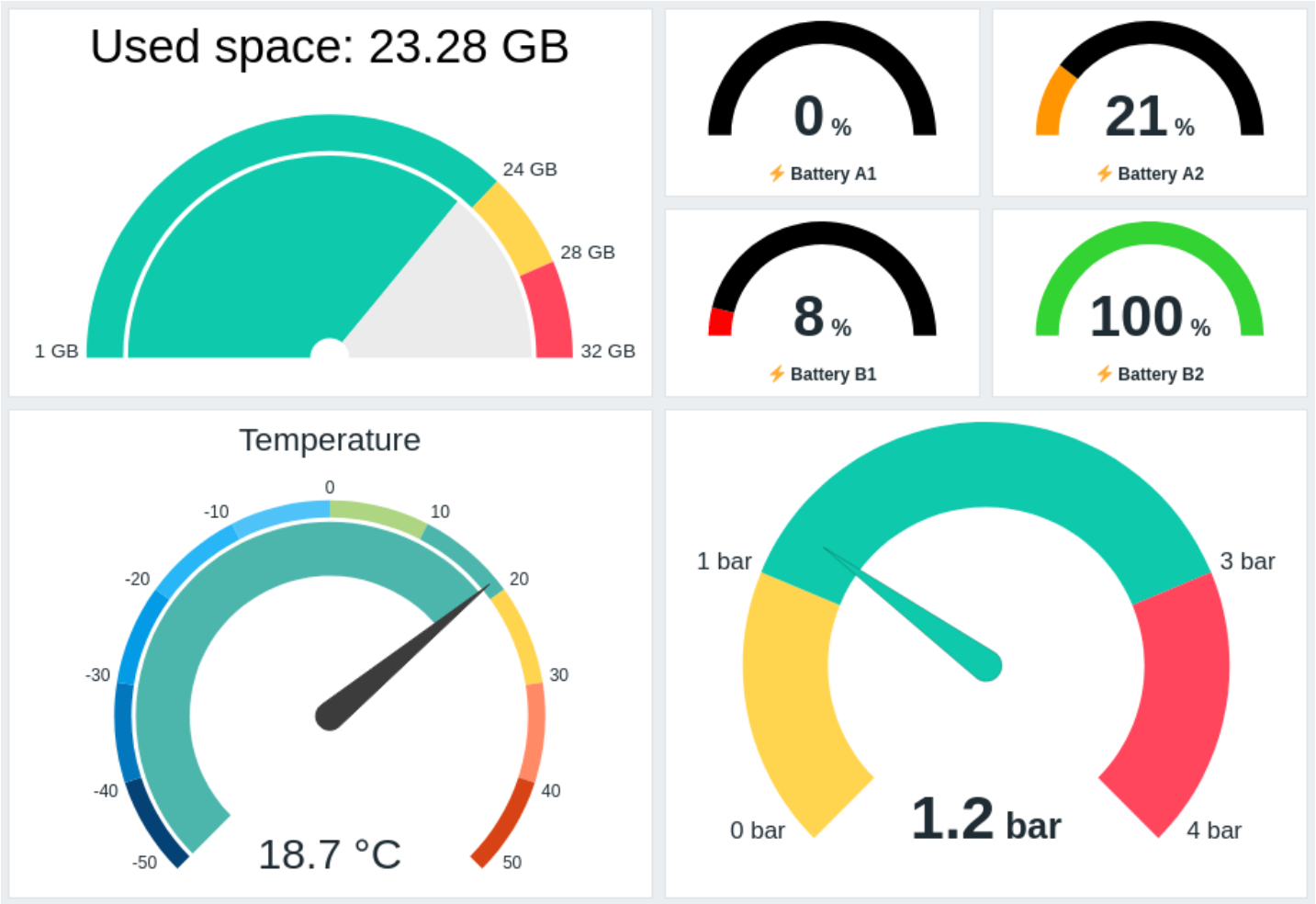
Charts used: Card

1. Card for single value / Multi line Card for multiple values
2. Negative trending values are shown in Red. Nominal values are shown in Blue and Up trending values are show in Green

Example KPI dashboard



Charts Used: Gauge



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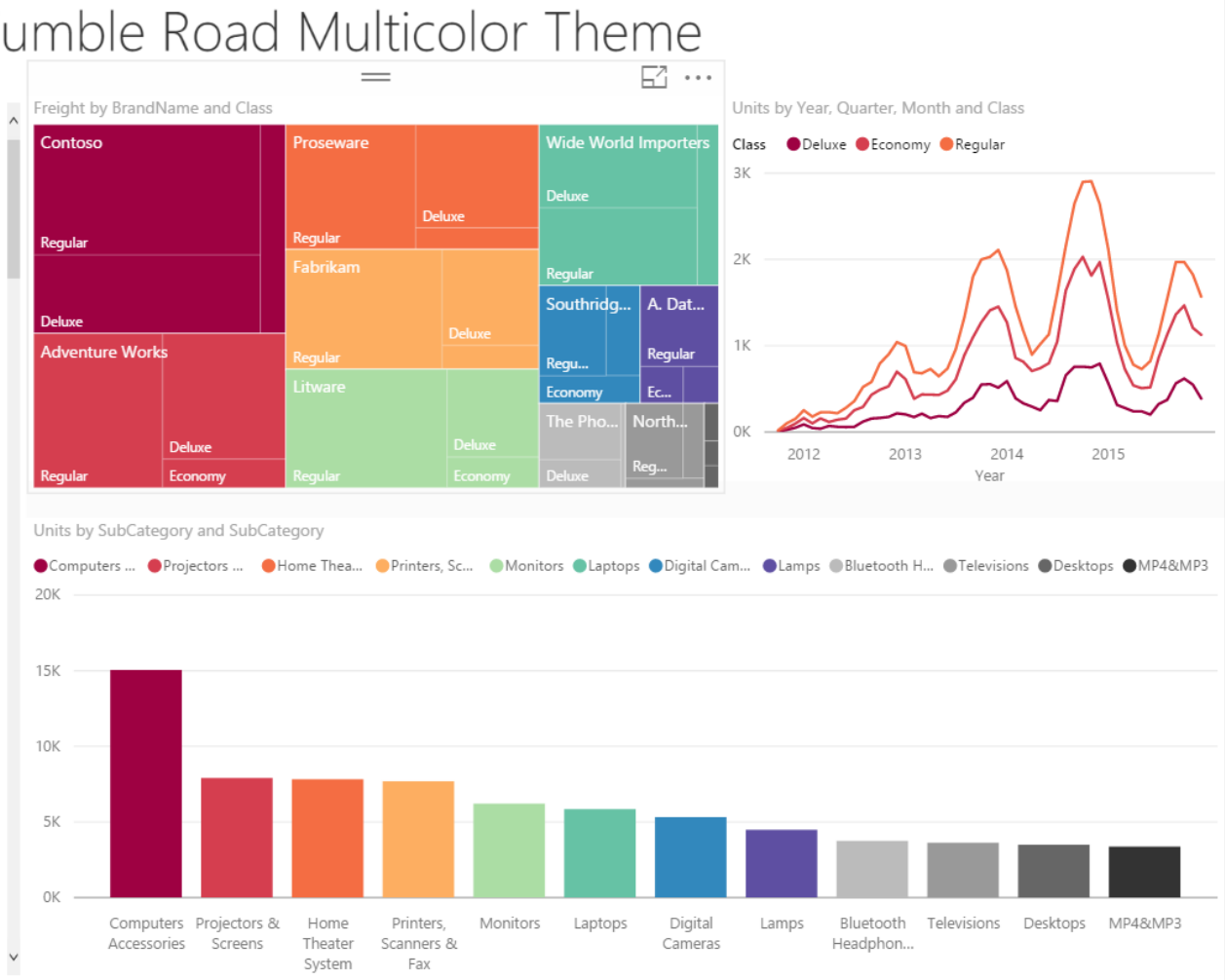
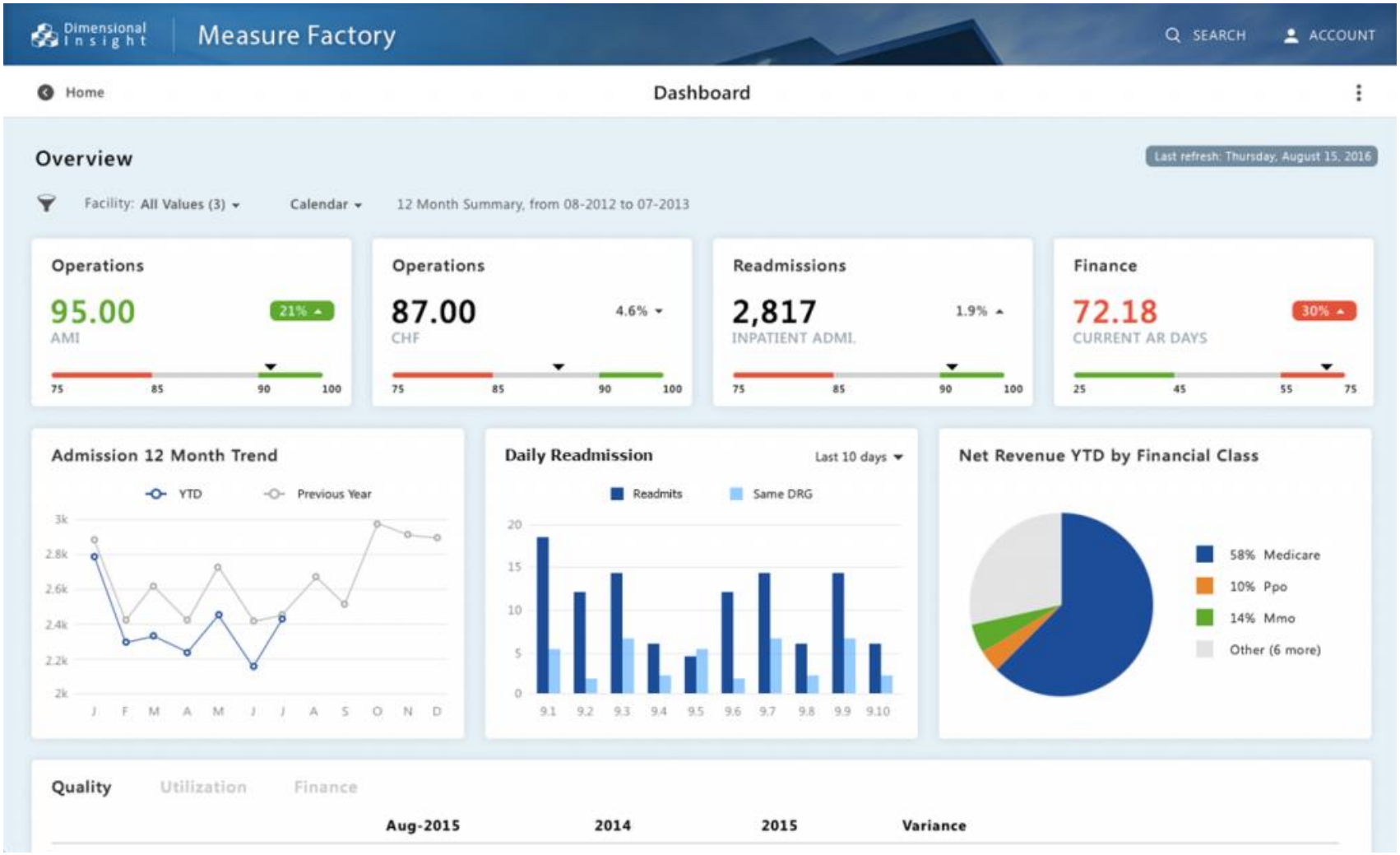
Study these…

Example 0

BrandName	Color	SalesAmount
A. Datum	Azure	67,346.20
A. Datum	Black	297,165.70
A. Datum	Blue	50,678.00
A. Datum	Gold	62,037.00
A. Datum	Green	71,799.10
A. Datum	Grey	277,681.00
A. Datum	Orange	126,412.20
A. Datum	Pink	242,893.40
A. Datum	Silver	246,452.30
A. Datum	Silver Grey	58,326.00
Adventure Works	Black	1,768,682.13
Adventure Works	Blue	221,448.24
Adventure Works	Brown	886,528.48
Adventure Works	Grey	146,383.73
Adventure Works	Red	251,808.15
Adventure Works	Silver	1,340,762.78
Adventure Works	White	1,715,546.01
Contoso	Black	1,684,896.31
Total		**40,640,973.42**



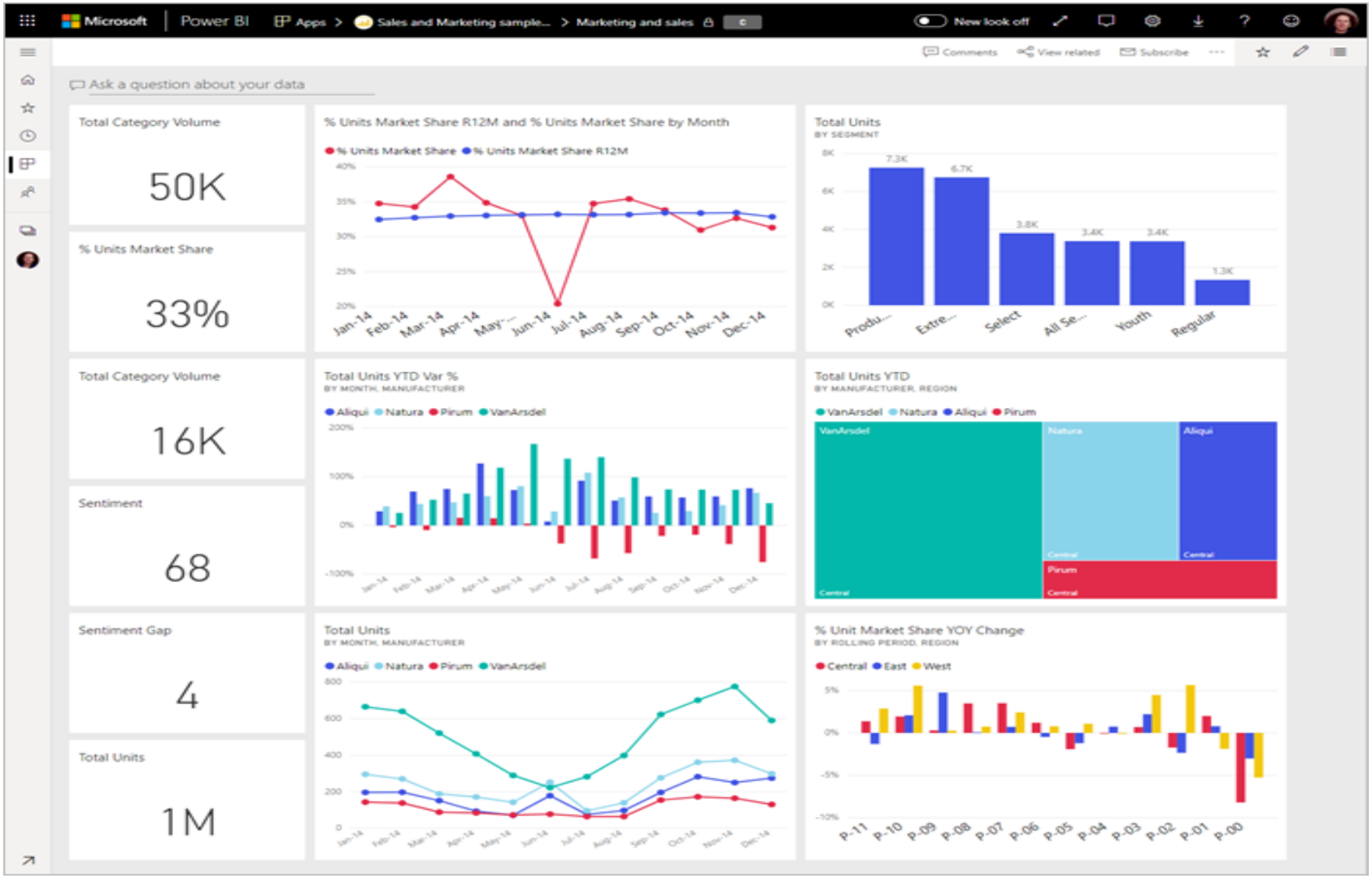
Example 1


Dimensional Insight
Measure Factory
SEARCH
ACCOUNT
Home
Dashboard
Overview
Last refresh: Thursday, August 15, 2016
Facility: All Values (3)
Calendar
12 Month Summary, from 08-2012 to 07-2013
Operations
95.00
AMI
21%
75 85 90 100
Operations
87.00
CHF
4.6%
75 85 90 100
Readmissions
2,817
INPATIENT ADMI.
1.9%
75 85 90 100
Finance
72.18
CURRENT AR DAYS
30%
25 45 55 75
Admission 12 Month Trend
YTD
Previous Year
3k 2.8k 2.6k 2.4k 2.2k 2k
J F M A M J J A S O N D
Daily Readmission
Last 10 days
Readmits
Same DRG
20 15 10 5 0
9.1 9.2 9.3 9.4 9.5 9.6 9.7 9.8 9.9 9.10
Net Revenue YTD by Financial Class
58% Medicare
10% Ppo
14% Mmo
Other (6 more)
Quality
Utilization
Finance
Aug-2015
2014
2015
Variance

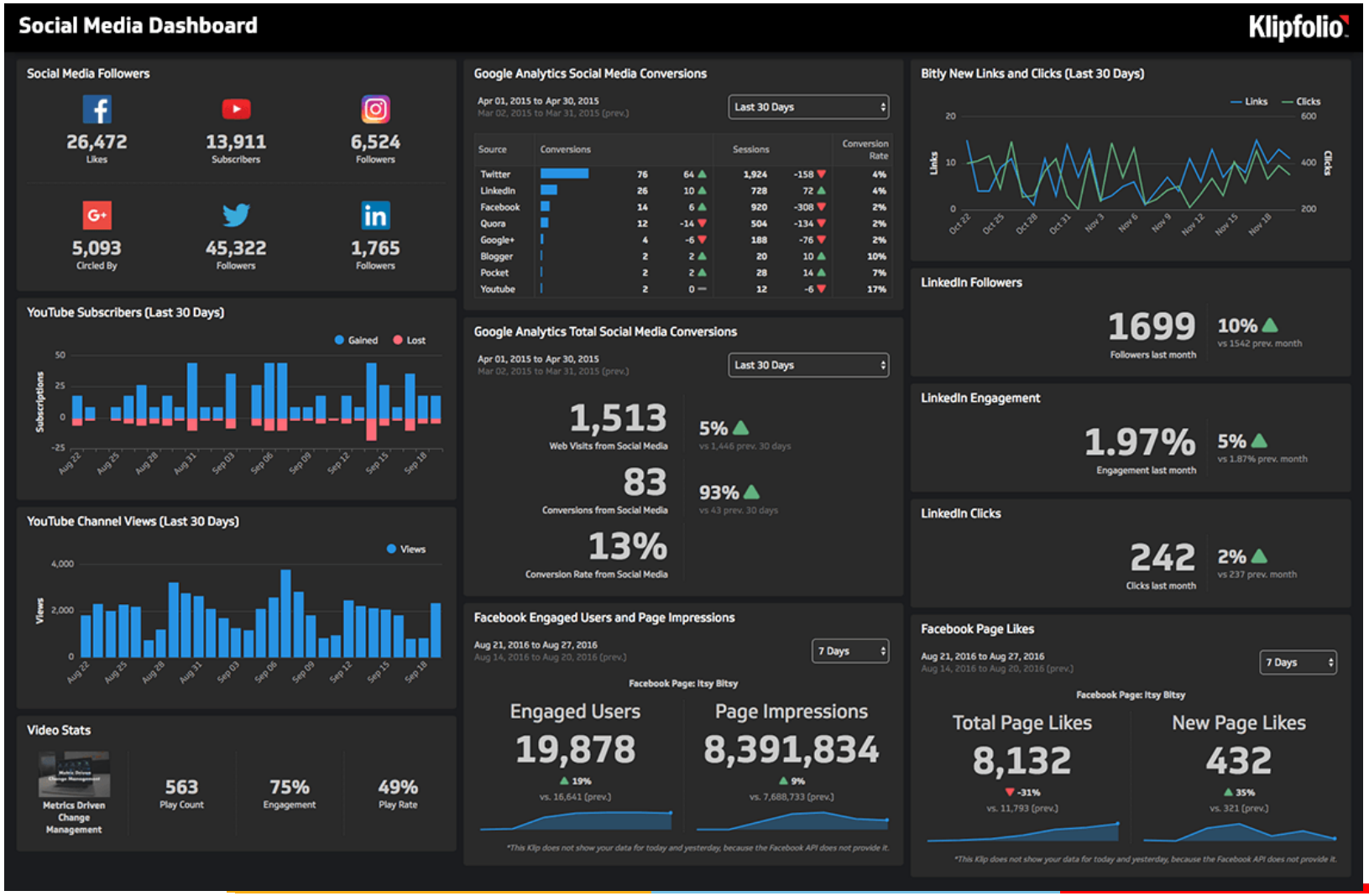
Example 2

1.029.973
VISITS
94 sec
AVG. SESSION DURATION
1,8 pages
PER VISIT
58 %
BOUNCE RATE
1.853.672
PAGE VIEWS
10%
GOAL CONVERSION
Visits by Week of Year
Bounce Rate by Week of Year
Traffic Sources
26 % Direct
5 % Display
58 % Organic
10 % Paid
5 % Referral
7 % Social
Visitors by User Type
69 %
31 %
New
Returning
Top 3 Channels by Conversion
Organic
17 %
Paid Search
11 %
Direct
10 %
Top 3 Campaigns by Conversion
Campaign 2
17 %
Campaign 3
10 %
Campaign 1
9 %
Top 3 Pages by Conversion
Page 2
19 %
Page 3
11 %
Page 1
11 %

Example 3


Microsoft
Power BI
Apps
Sales and Marketing sample...
Marketing and sales
New look off
Comments
View related
Subscribe
Ask a question about your data
Total Category Volume
50K
% Units Market Share
33%
Total Category Volume
16K
Sentiment
68
Sentiment Gap
4
Total Units
1M
% Units Market Share R12M and % Units Market Share by Month
% Units Market Share
% Units Market Share R12M
Total Units
BY SEGMENT
7.3K
6.7K
3.8K
3.4K
3.4K
1.3K
Produ...
Extre...
Select
All Se...
Youth
Regular
Total Units YTD Var %
BY MONTH, MANUFACTURER
Aliqui
Natura
Pirum
VanArsdel
Total Units YTD
BY MANUFACTURER, REGION
VanArsdel
Natura
Aliqui
Pirum
Central
Total Units
BY MONTH, MANUFACTURER
Aliqui
Natura
Pirum
VanArsdel
% Unit Market Share YOY Change
BY ROLLING PERIOD, REGION
Central
East
West
P-11
P-10
P-09
P-08
P-07
P-06
P-05
P-04
P-03
P-02
P-01
P-00

Example 4


Social Media Dashboard
Klipfolio
Social Media Followers
26,472 Likes
13,911 Subscribers
6,524 Followers
5,093 Circled By
45,322 Followers
1,765 Followers
YouTube Subscribers (Last 30 Days)
Gained
Lost
Subscriptions
YouTube Channel Views (Last 30 Days)
Views
Video Stats
Metrics Driven Change Management
563 Play Count
75% Engagement
49% Play Rate
Google Analytics Social Media Conversions
Apr 01, 2015 to Apr 30, 2015
Mar 02, 2015 to Mar 31, 2015 (prev.)
Last 30 Days
Source
Conversions
Sessions
Conversion Rate
Twitter 76 64 1,924 -158 4%
LinkedIn 26 10 728 72 4%
Facebook 14 6 920 -308 2%
Quora 12 -14 504 -134 2%
Google+ 4 -6 188 -76 2%
Blogger 2 2 20 10 10%
Pocket 2 2 28 14 7%
Youtube 2 0 12 -6 17%
Google Analytics Total Social Media Conversions
Apr 01, 2015 to Apr 30, 2015
Mar 02, 2015 to Mar 31, 2015 (prev.)
Last 30 Days
1,513 Web Visits from Social Media
5% vs 1,446 prev. 30 days
83 Conversions from Social Media
93% vs 43 prev. 30 days
13% Conversion Rate from Social Media
Facebook Engaged Users and Page Impressions
Aug 21, 2016 to Aug 27, 2016
Aug 14, 2016 to Aug 20, 2016 (prev.)
7 Days
Facebook Page: Itsy Bitsy
Engaged Users
19,878
19%
vs. 16,641 (prev.)
Page Impressions
8,391,834
9%
vs. 7,688,733 (prev.)
*This Klip does not show your data for today and yesterday, because the Facebook API does not provide it.
Bitly New Links and Clicks (Last 30 Days)
Links
Clicks
LinkedIn Followers
1699 Followers last month
10% vs 1542 prev. month
LinkedIn Engagement
1.97% Engagement last month
5% vs 1.87% prev. month
LinkedIn Clicks
242 Clicks last month
2% vs 237 prev. month
Facebook Page Likes
Aug 21, 2016 to Aug 27, 2016
Aug 14, 2016 to Aug 20, 2016 (prev.)
7 Days
Facebook Page: Itsy Bitsy
Total Page Likes
8,132
-31%
vs. 11,793 (prev.)
New Page Likes
432
35%
vs. 321 (prev.)
*This Klip does not show your data for today and yesterday, because the Facebook API does not provide it.

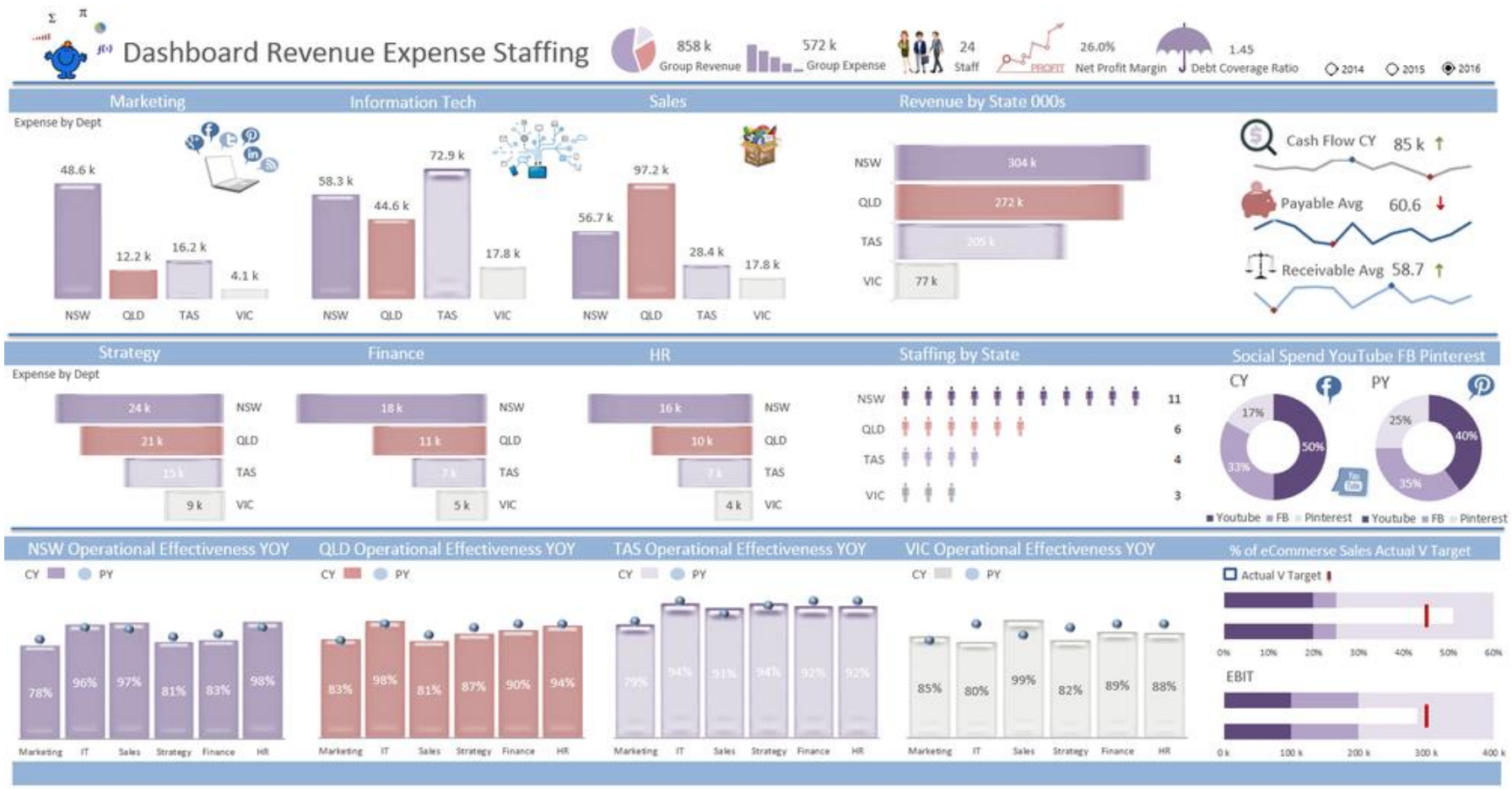
Example 5

Cyfe
Marketing Dashboard
Jan 1, 2012 - Jan 31, 2012
AUDIENCE OVERVIEW
404 ▲71%
PAGEVIEWS
265 ▲96%
VISITS
264 ▲98%
VISITORS
Jan 2
Jan 9
Jan 16
Jan 23
Jan 30
Google Analytics
WORLD HEAT MAP
Google Analytics
CONTENT OVERVIEW
00:00:28 ▲13%
TIME ON SITE
61.07% ▼3%
BOUNCE RATE
65.54%
EXIT RATE
Jan 2
Jan 9
Jan 16
Jan 23
Jan 30
Google Analytics
TRAFFIC OVERVIEW
Campaign (0%)
Direct (33%)
Search (59%)
Referral (8%)
Google Analytics
generic logo
company
Image
CONVERSION OVERVIEW
1,226
CONVERSIONS
3.43%
CONVERSION RATE
Jan 2
Jan 9
Jan 16
Jan 23
Jan 30
Google Analytics
TOP TRAFFIC SOURCES
SOURCE
VISITS
1. google / organic 151 ▲53%
2. (direct) / (none) 87 ▲1350%
3. www.forex-ninjas.com / referral 14
4. forex-ninjas.com / referral 8
5. yahoo / organic 4 ▲33%
Google Analytics
TOP BROWSER TYPES
BROWSER
VISITS
1. Safari 104 ▲447%
2. Chrome 64 ▲113%
3. Firefox 55 ▲34%
4. Internet Explorer 29 ▼9%
5. Opera 5 ▲400%
6. Mozilla Compatible Agent 4 ▲100%
7. Opera 9.4 2
8. IE with Chrome Frame 1
Google Analytics
TOP CONVERSION SOURCES
SOURCE
CONV. RATE
1. google.es 57.1
2. readwriteweb.com 45.1
3. wwwhatsnew.com 38.5
4. br.wwwhatsnew.com 38.1
5. feedly.com 36.4
Google Analytics
DARK

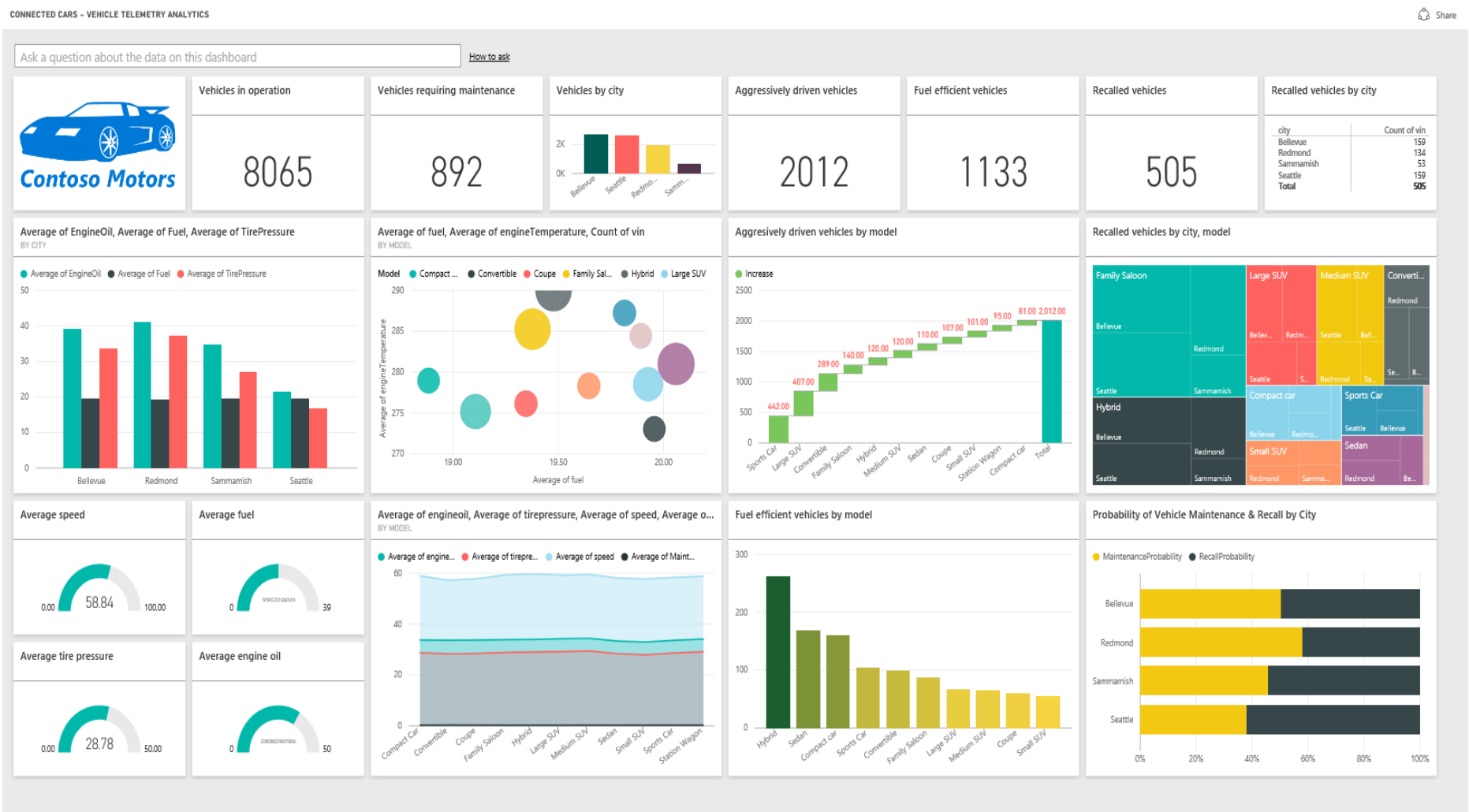
Example 6

Financial Trends by Region
Revenue Plan Actual Var Trend
England 15.3 m 16.2 m 6.1%
France 18.7 m 19.1 m 1.9%
Germany 18.5 m 17.2 m (7.0%)
Greece 26.3 m 25.6 m (2.6%)
Ireland 17.7 m 18.4 m 4.0%
Italy 6.0 m 8.6 m 43.6%
Portugal 15.1 m 14.5 m (4.3%)
Scotland 10.7 m 11.4 m 6.0%
Spain 29.8 m 23.9 m (19.8%)
Wales 9.2 m 9.8 m 6.1%
Africa
Americas
Asia Pac
Europe
Expense Plan Actual Var Trend
England 5.7 m 6.1 m 6.1%
France 7.1 m 6.5 m (8.0%)
Germany 6.9 m 6.6 m (4.3%)
Greece 9.9 m 10.5 m 6.1%
Ireland 6.7 m 7.1 m 6.2%
Italy 2.2 m 4.2 m 88.1%
Portugal 5.7 m 5.4 m (4.3%)
Scotland 4.0 m 4.3 m 6.0%
Spain 11.9 m 9.5 m (20.4%)
Wales 3.5 m 3.7 m 6.1%
April
370 Office
80% Efficiency
453 Staf
10.7 ROI
16.6% ROA
* Hold down the Control Key to Choose multiple Clients. Click on the Chart Series to update the chart.
$ 000s
Millions
Date
Turnover
England
Nireland
Holland
France
Greece
Russia
FTE Change - Trend
Bottom 5
Footfall Change
-1.6%
-0.8%
0.7%
1.3%
1.9%
Portugal
France
Greece
Germany
Ireland
Revenue and Expenses Trend
139
187
57
Re Sellers
Products
Lines
Company
Denmark
England
France
Germany
Greece
Holland
Italy
Nireland
Portugal
Russia
Scotland
Spain
Sweeden
Turkey
Wales
Actual
Plan
Projects Actual v Plan
Efficiency Trend by Region
Market Share
Market Cap
Market Share and Market Cap by Month

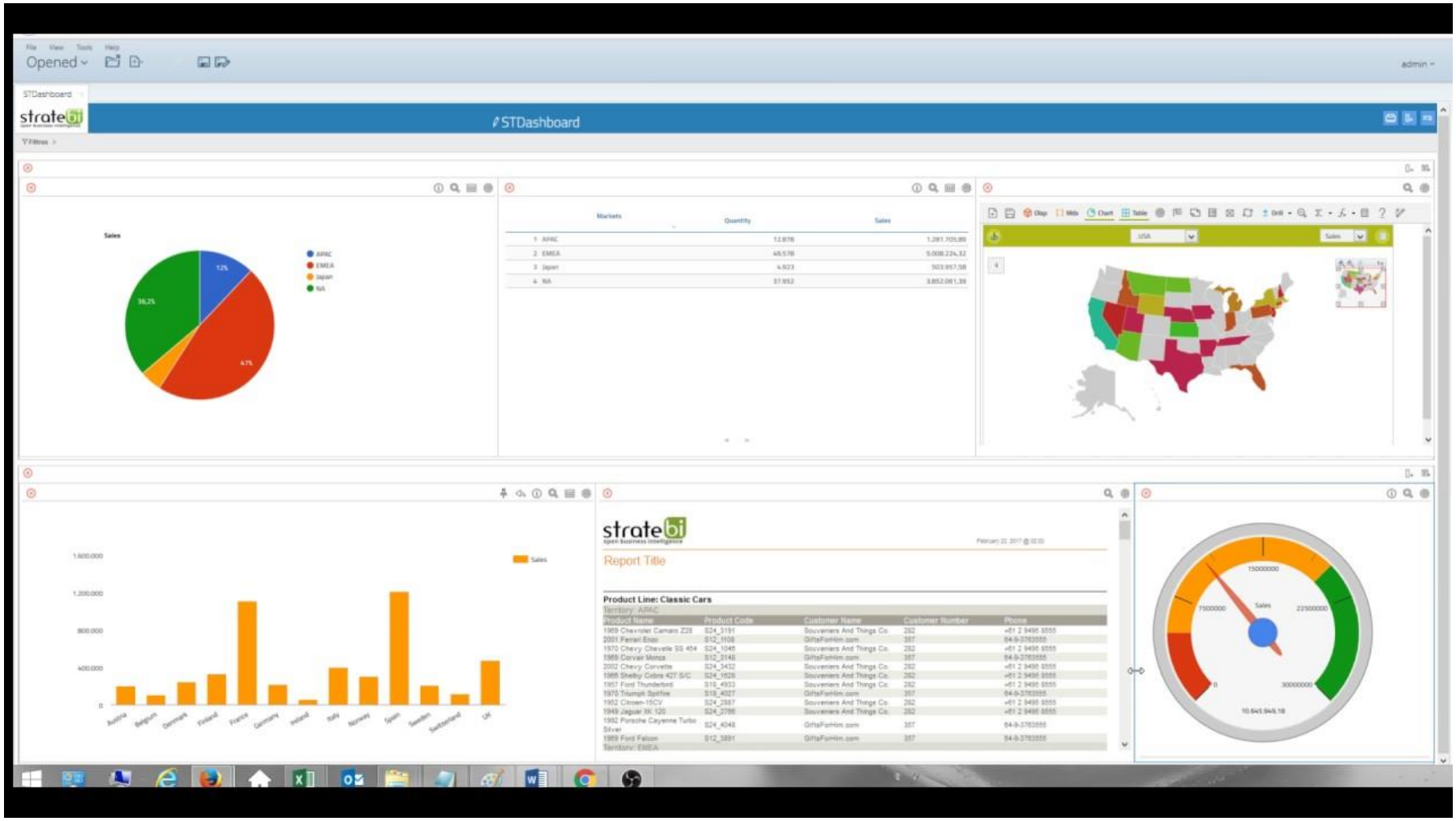
Example 7


Dashboard Revenue Expense Staffing
858 k
Group Revenue
572 k
Group Expense
24
Staff
26.0%
Net Profit Margin
1.45
Debt Coverage Ratio
2014
2015
2016
Marketing
Expense by Dept
48.6 k
12.2 k
16.2 k
4.1 k
NSW
QLD
TAS
VIC
Information Tech
58.3 k
44.6 k
72.9 k
17.8 k
Sales
56.7 k
97.2 k
28.4 k
17.8 k
Revenue by State 000s
NSW
304 k
QLD
272 k
TAS
VIC
77 k
Cash Flow CY
85 k
Payable Avg
60.6
Receivable Avg
58.7
Strategy
Expense by Dept
24 k
21 k
9 k
Finance
18 k
11 k
5 k
HR
16 k
10 k
4 k
Staffing by State
11
6
4
3
Social Spend YouTube FB Pinterest
CY
PY
17%
50%
33%
25%
40%
35%
Youtube
FB
Pinterest
NSW Operational Effectiveness YOY
QLD Operational Effectiveness YOY
TAS Operational Effectiveness YOY
VIC Operational Effectiveness YOY
CY
PY
78%
96%
97%
81%
83%
98%
83%
98%
81%
87%
90%
94%
85%
80%
99%
82%
89%
88%
Marketing
IT
Sales
Strategy
Finance
HR
% of eCommerse Sales Actual V Target
Actual V Target
EBIT

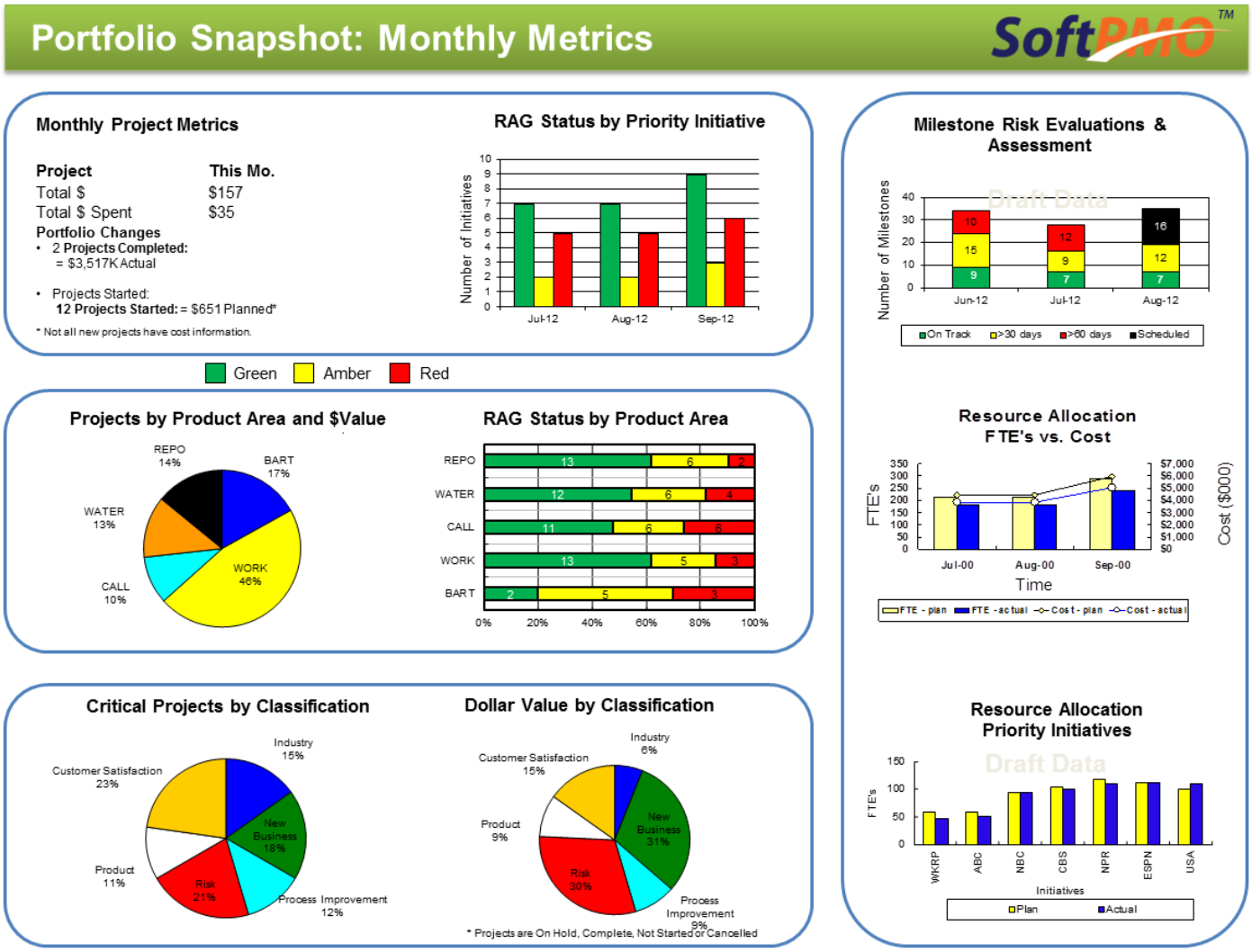
Example 8


CONNECTED CARS - VEHICLE TELEMETRY ANALYTICS
Share
Ask a question about the data on this dashboard
How to ask
Contoso Motors
Vehicles in operation
8065
Vehicles requiring maintenance
892
Vehicles by city
Aggressively driven vehicles
2012
Fuel efficient vehicles
1133
Recalled vehicles
505
Recalled vehicles by city
city
Count of vin
Bellevue 159
Redmond 134
Sammamish 53
Seattle 159
Total 505
Average of EngineOil, Average of Fuel, Average of TirePressure
BY CITY
Average of EngineOil
Average of Fuel
Average of TirePressure
Bellevue
Redmond
Sammamish
Seattle
Average of fuel, Average of engineTemperature, Count of vin
BY MODEL
Model
Compact ...
Convertible
Coupe
Family Sal...
Hybrid
Large SUV
Average of engineTemperature
Average of fuel
Aggresively driven vehicles by model
Increase
Sports Car
Large SUV
Convertible
Family Saloon
Hybrid
Medium SUV
Sedan
Coupe
Small SUV
Station Wagon
Compact car
Total
Recalled vehicles by city, model
Family Saloon
Large SUV
Medium SUV
Hybrid
Compact car
Sports Car
Small SUV
Sedan
Average speed
58.84
Average fuel
Average tire pressure
28.78
Average engine oil
Average of engineoil, Average of tirepressure, Average of speed, Average o...
BY MODEL
Average of engine...
Average of tirepre...
Average of speed
Average of Maint...
Fuel efficient vehicles by model
Probability of Vehicle Maintenance & Recall by City
MaintenanceProbability
RecallProbability

Example 9


Opened
STDashboard
stratebi
STDashboard
Sales
Markets
Quantity
Sales
Report Title
Product Line: Classic Cars

Example 10



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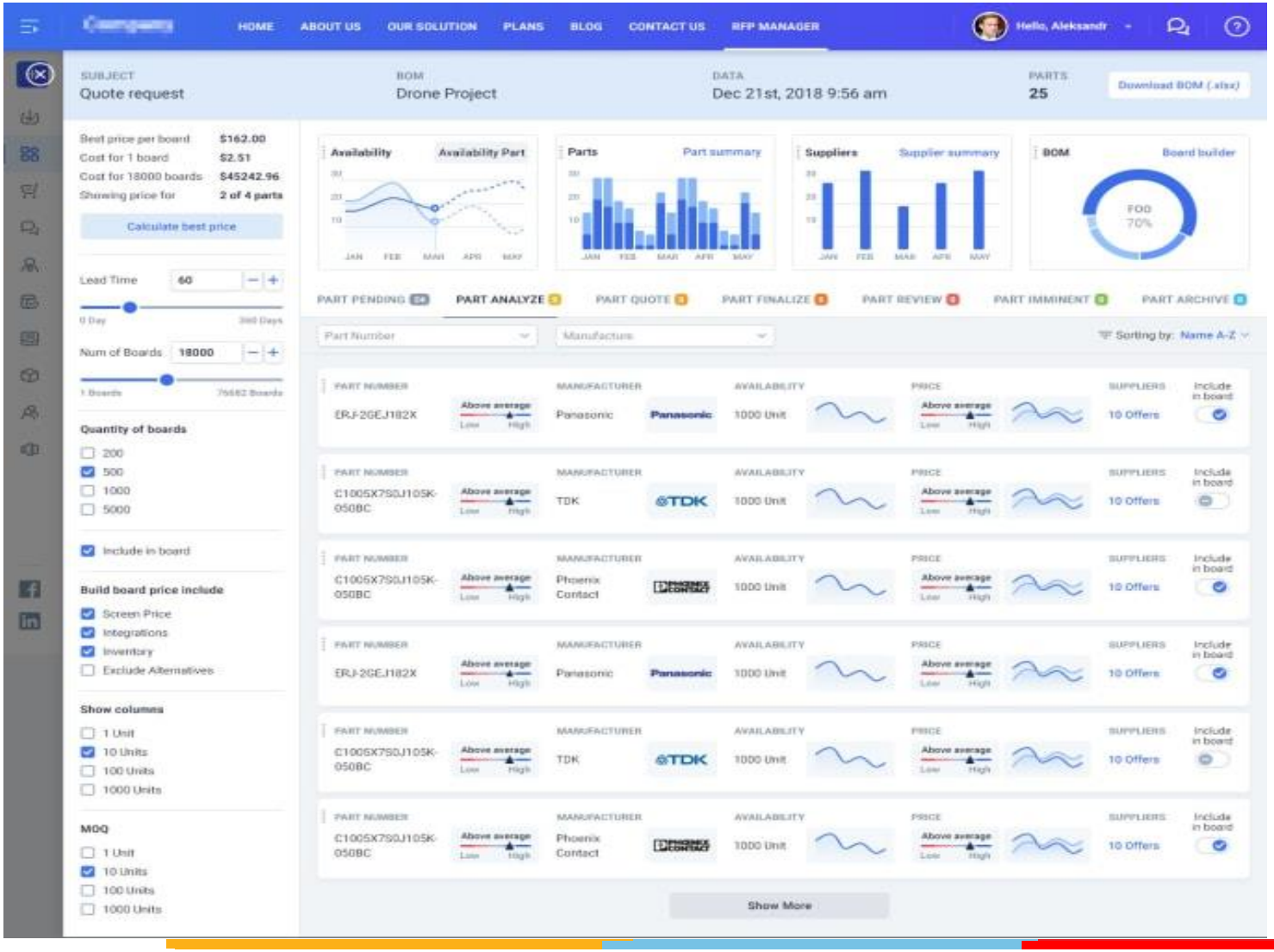
Complexxxx


HOME
ABOUT US
OUR SOLUTION
PLANS
BLOG
CONTACT US
RFP MANAGER
Hello, Aleksandr
SUBJECT
Quote request
BOM
Drone Project
DATA
Dec 21st, 2018 9:56 am
PARTS
25
Download BOM (.xlsx)
Best price per board $162.00
Cost for 1 board $2.51
Cost for 18000 boards $45242.96
Showing price for 2 of 4 parts
Calculate best price
Lead Time 60
Num of Boards 18000
Quantity of boards
200
500
1000
5000
Include in board
Build board price include
Screen Price
Integrations
Inventory
Exclude Alternatives
Show columns
1 Unit
10 Units
100 Units
1000 Units
MOQ
1 Unit
10 Units
100 Units
1000 Units
Availability
Availability Part
Parts
Part summary
Suppliers
Supplier summary
BOM
Board builder
FOO
70%
PART PENDING
PART ANALYZE
PART QUOTE
PART FINALIZE
PART REVIEW
PART IMMINENT
PART ARCHIVE
Part Number
Manufacture
Sorting by: Name A-Z
PART NUMBER
MANUFACTURER
AVAILABILITY
PRICE
SUPPLIERS
Include in board
ERJ-2GEJ182X
Panasonic
1000 Unit
10 Offers
C1005X7S0J105K-050BC
TDK
Phoenix Contact
Above average
Low
High
Show More

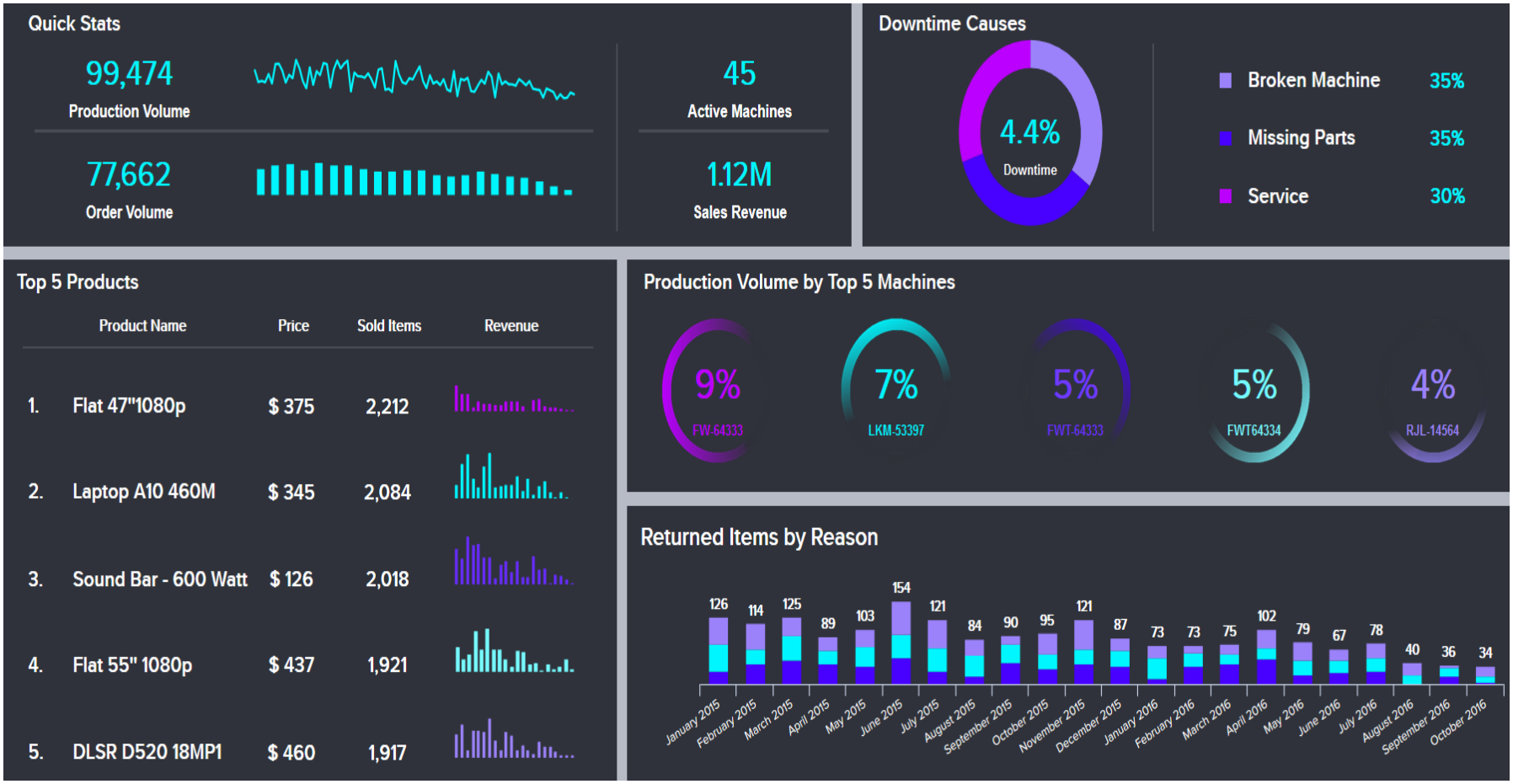
innovate achieve lead

BITS Pilani

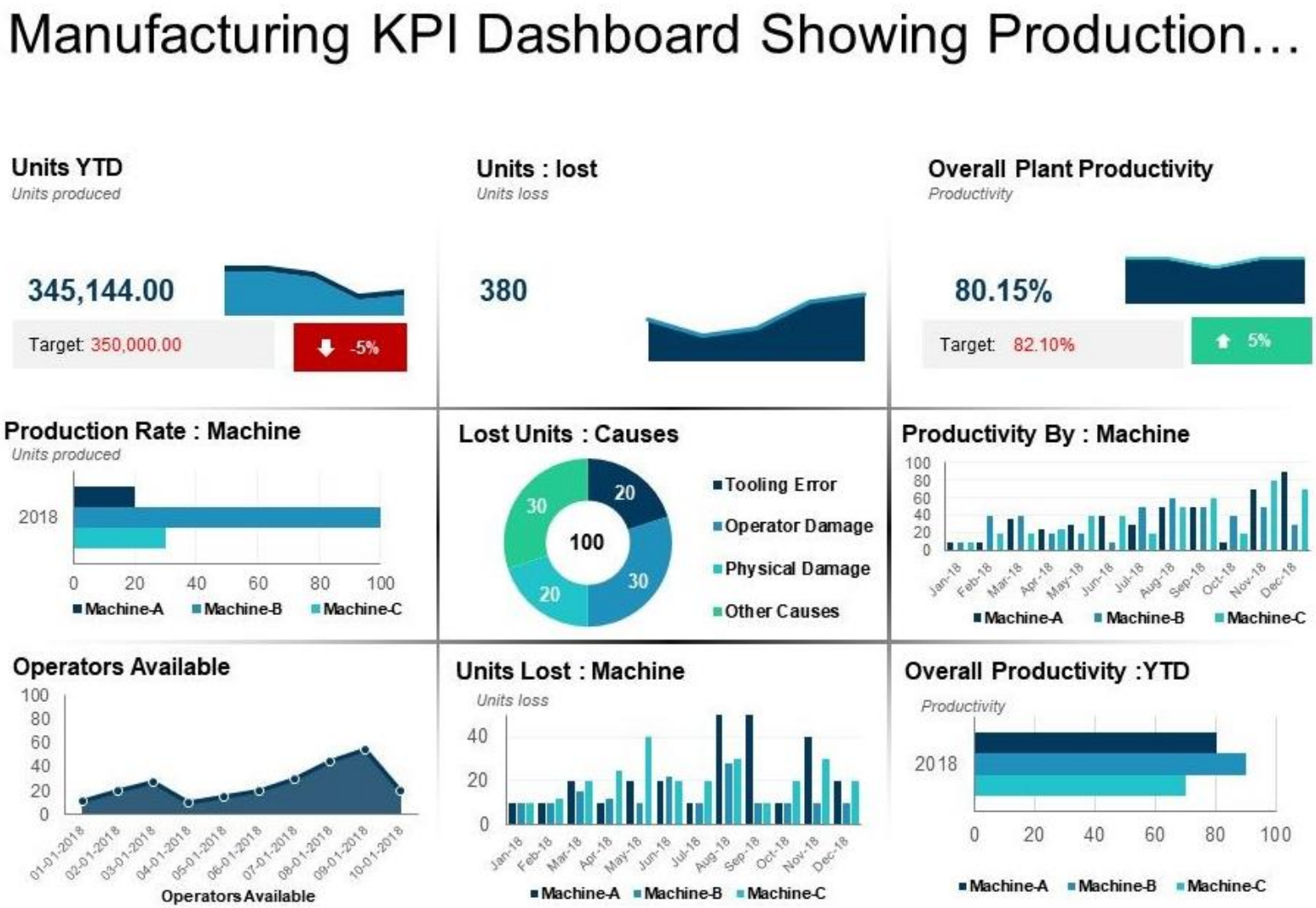
Pilani|Dubai|Goa|Hyderabad

Example Dashboards with KPIs

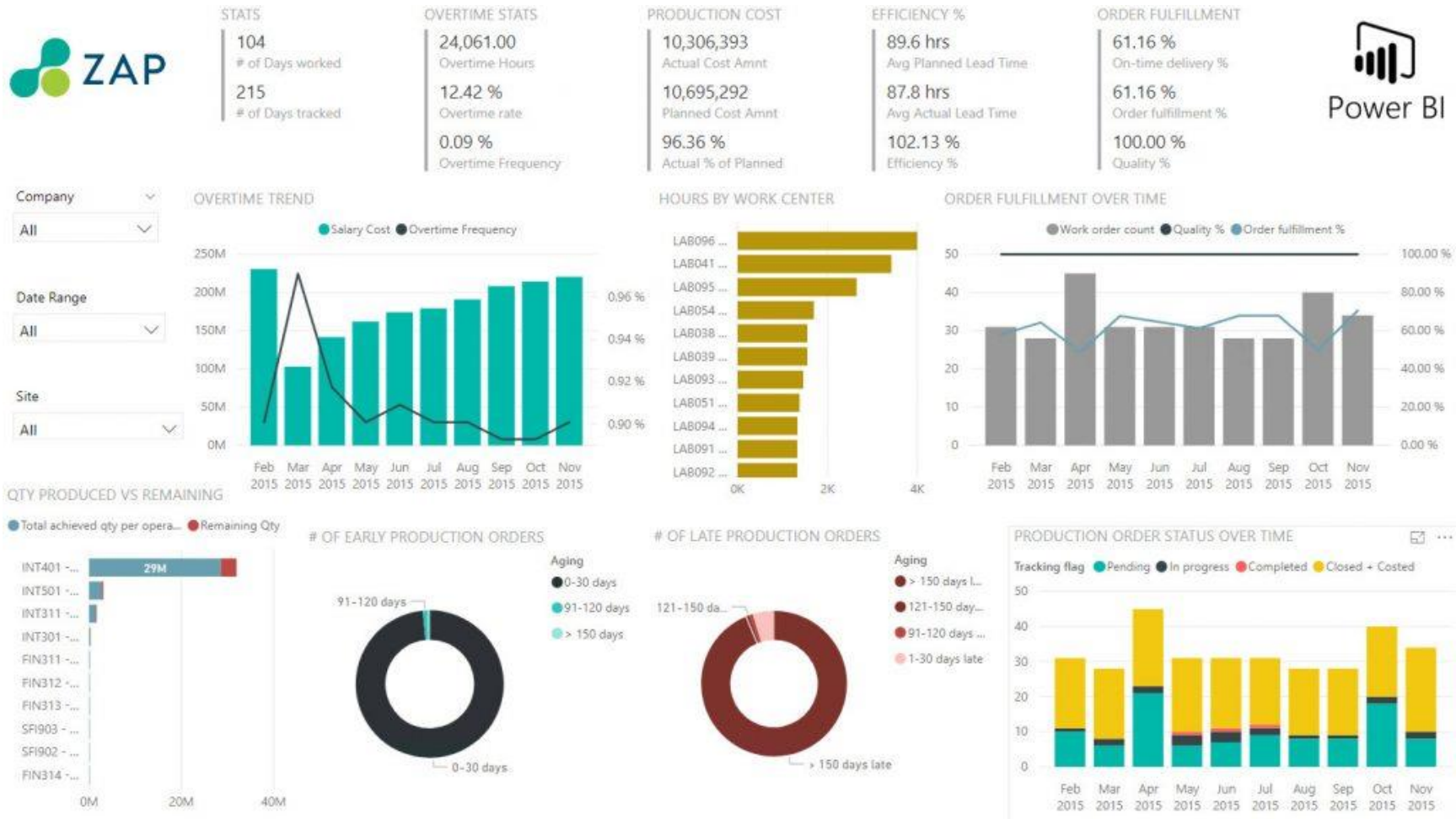
Manufacturing Dashboard



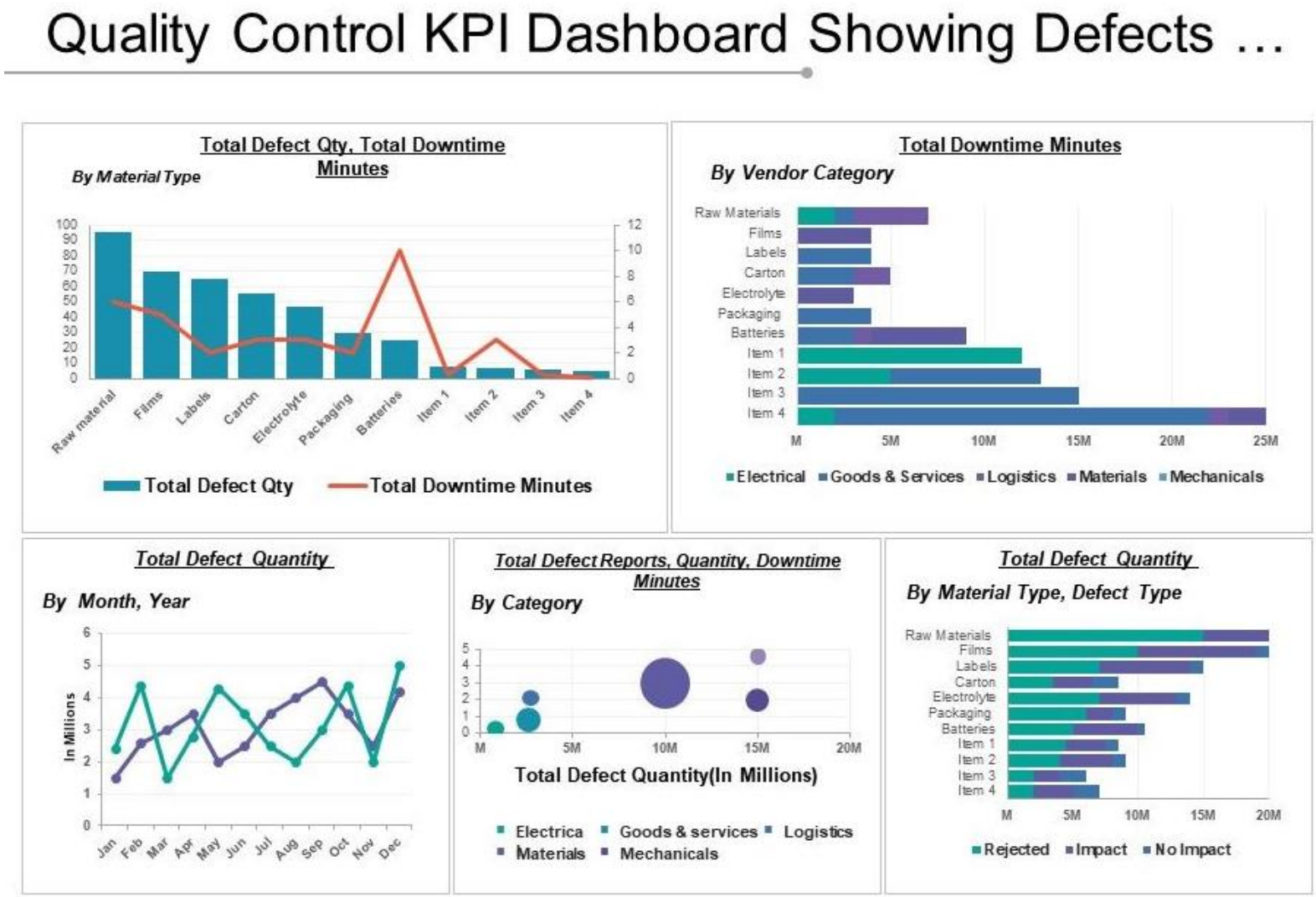
Manufacturing Dashboard (2)



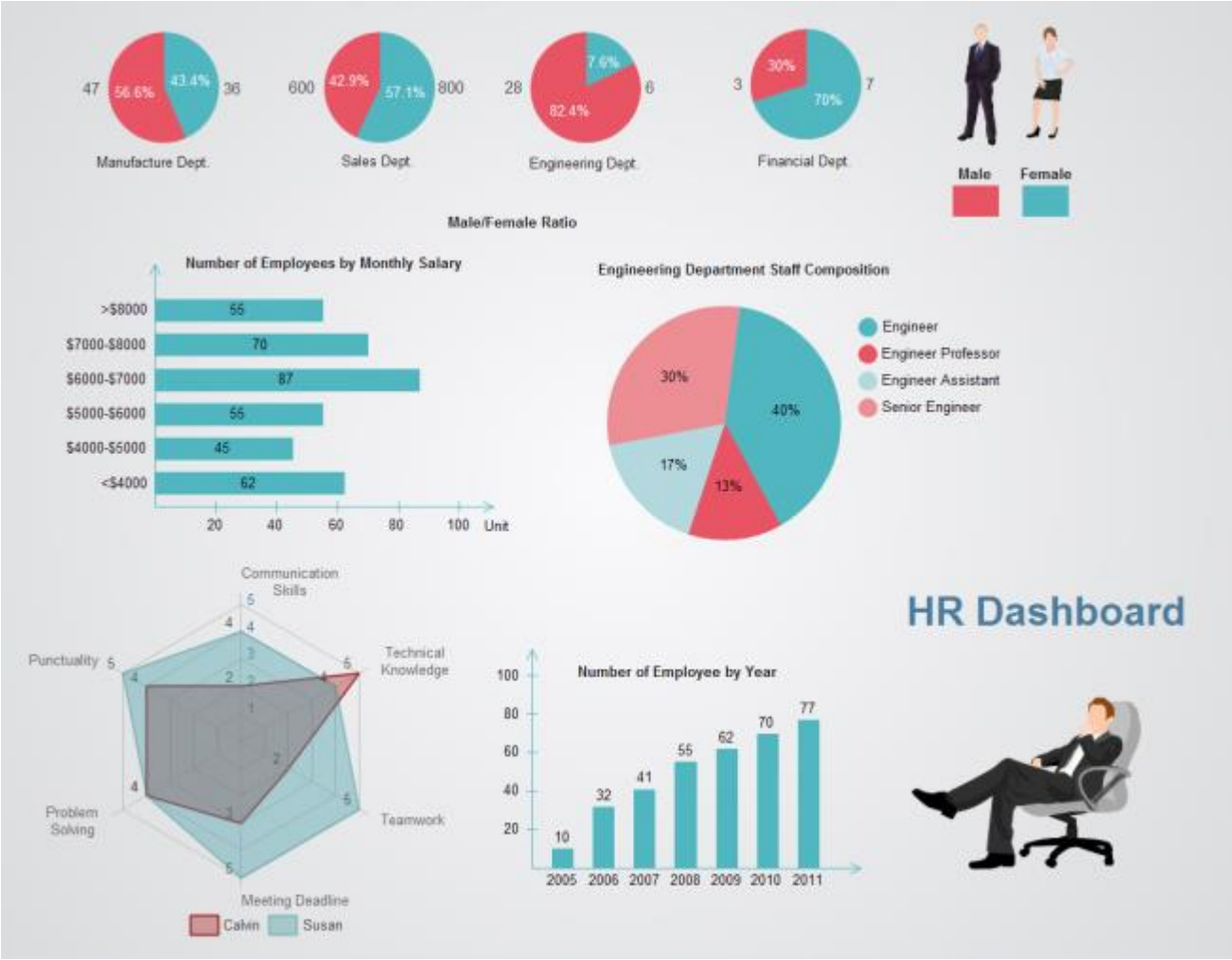
Production Dashboard (3)



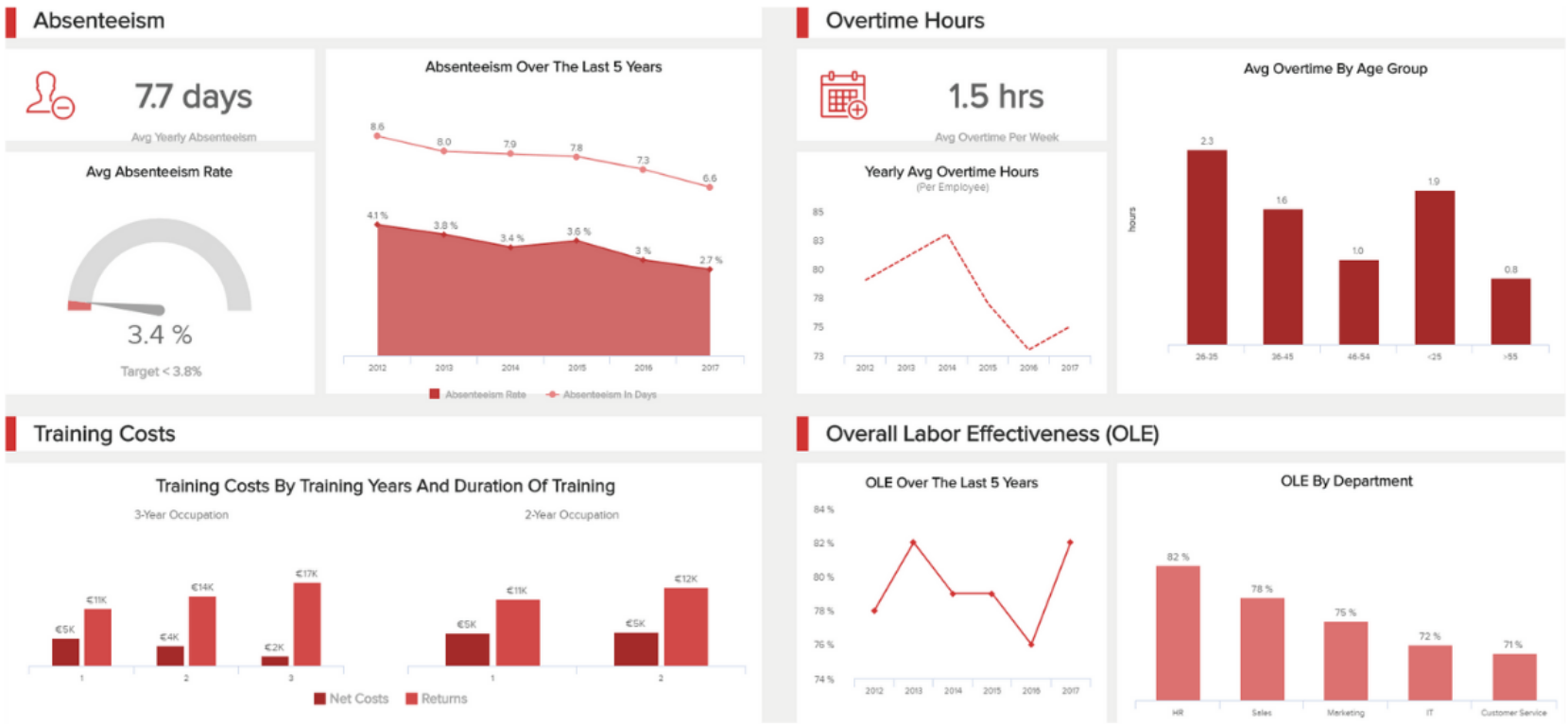
Quality Control Dashboard



HR Dashboard

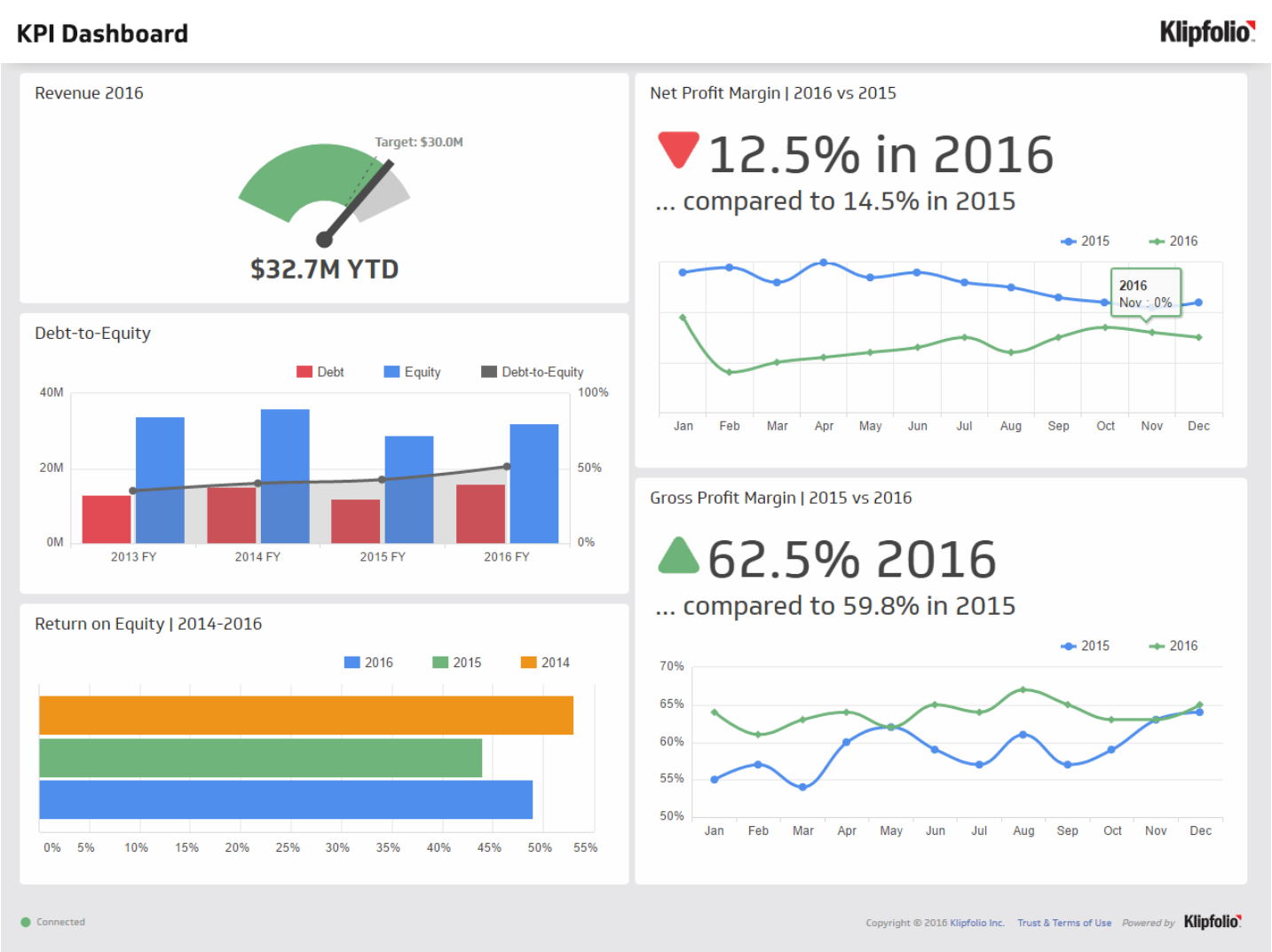


HR Dashboard (2)

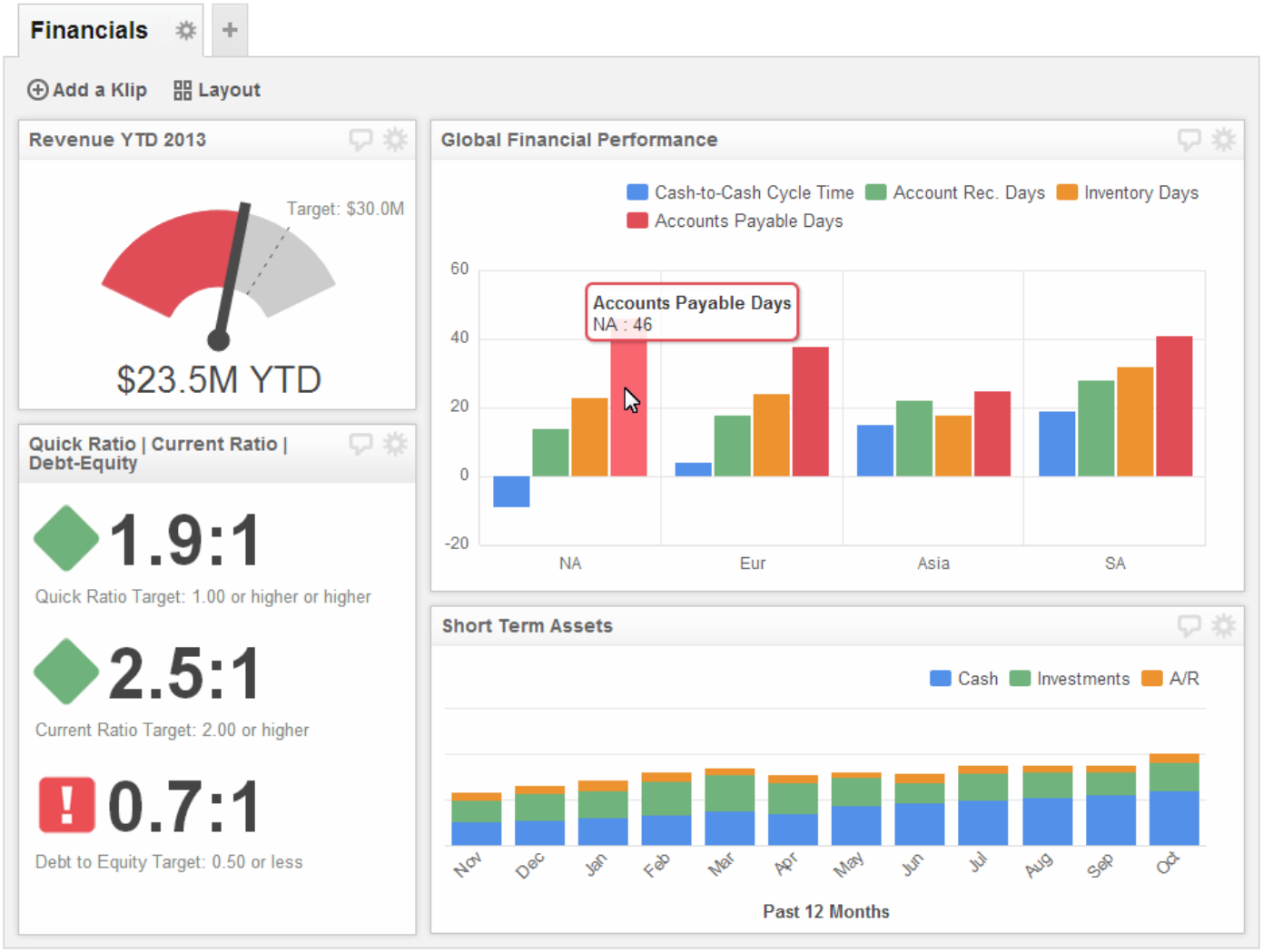


Energy Sector Dashboard

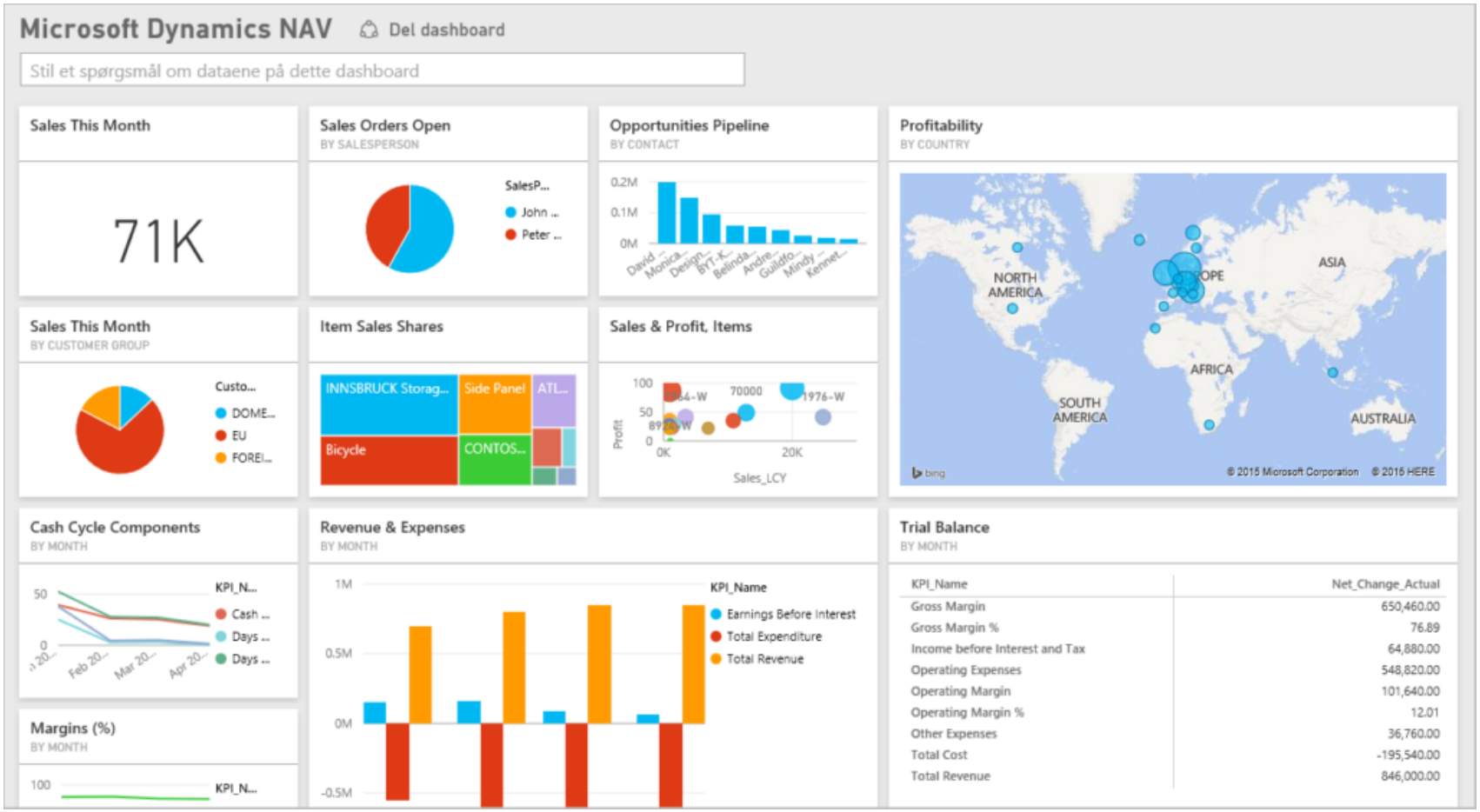
Finance KPI Dashboard



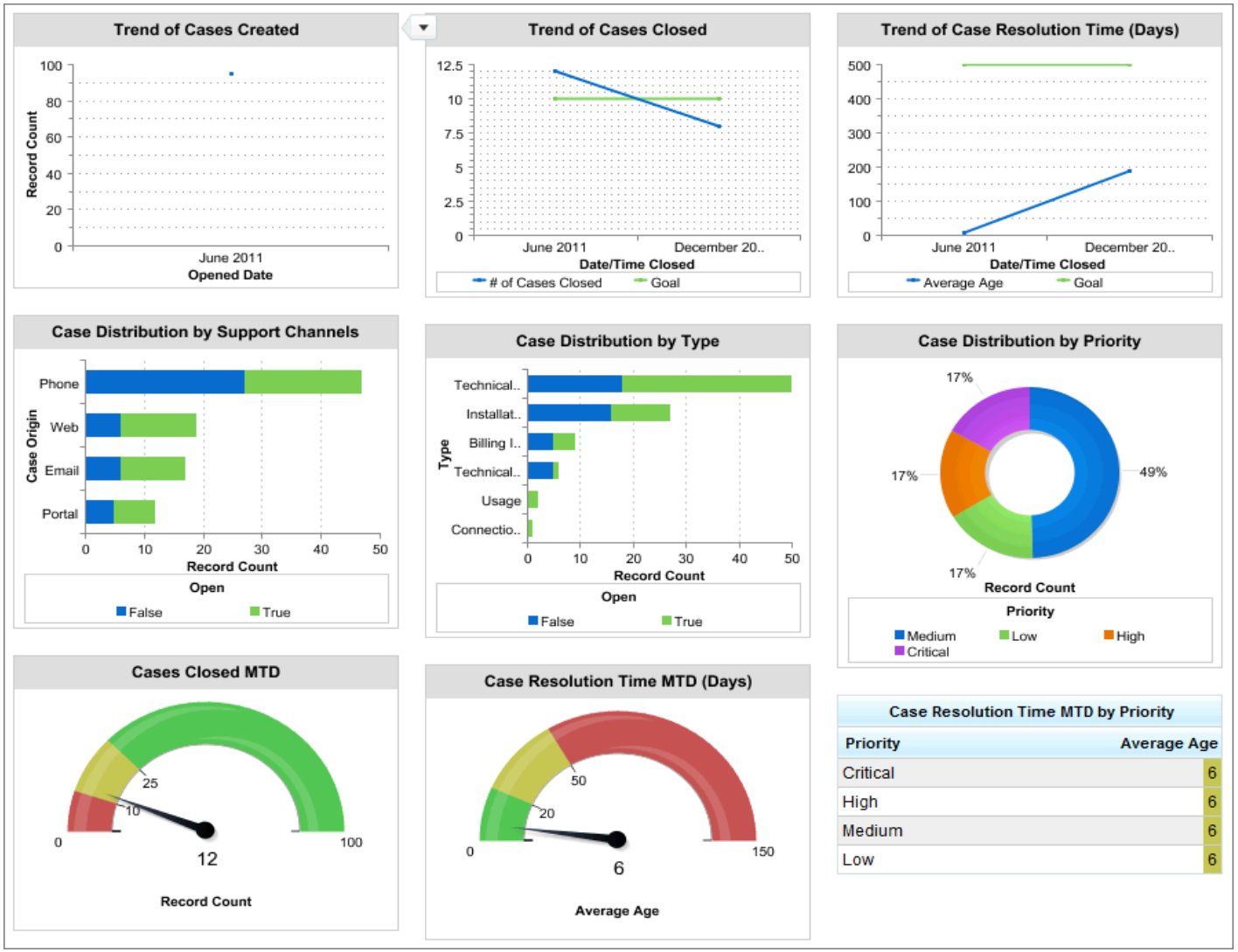
Financial Dashboard (2)



Spend analysis

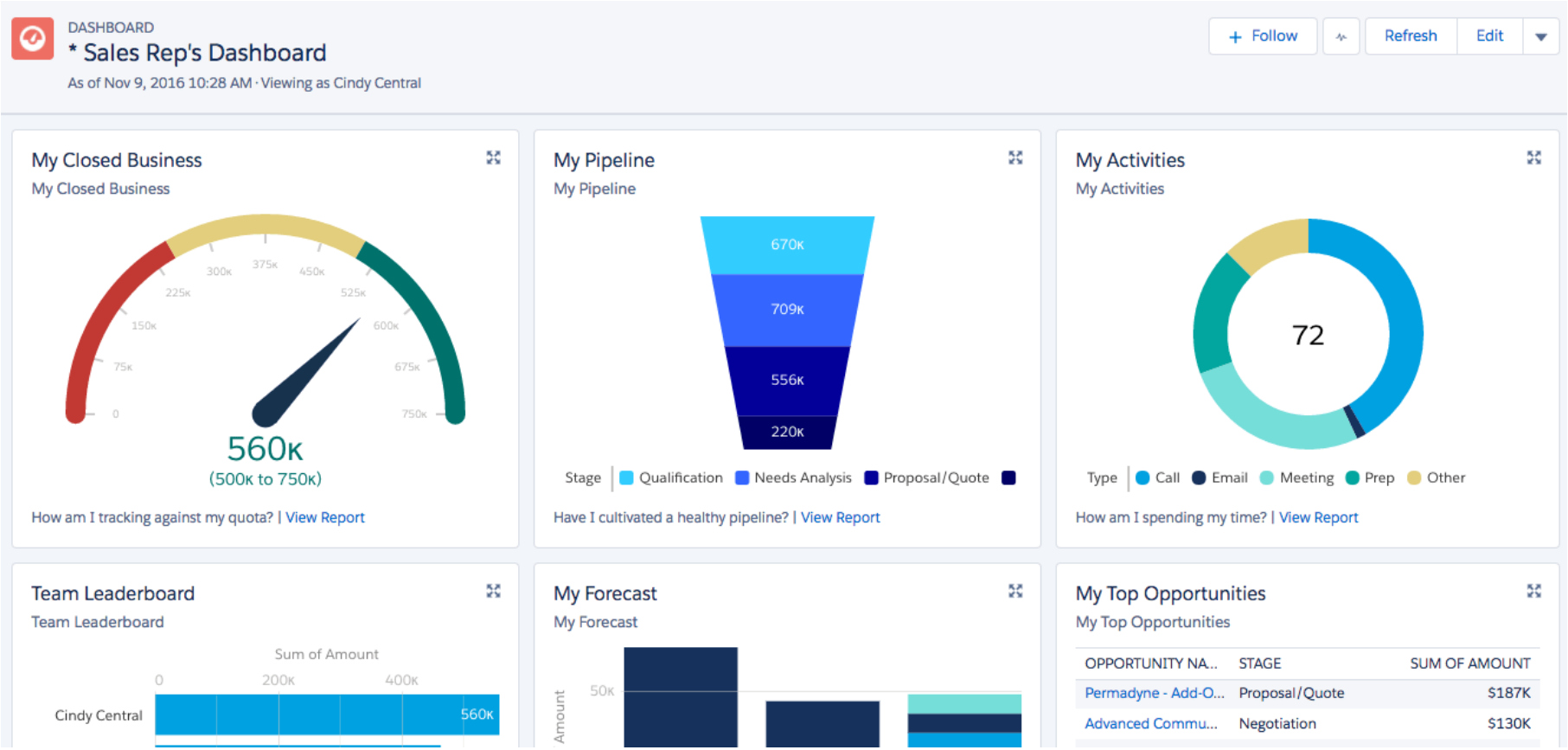


Customer Support KPIs

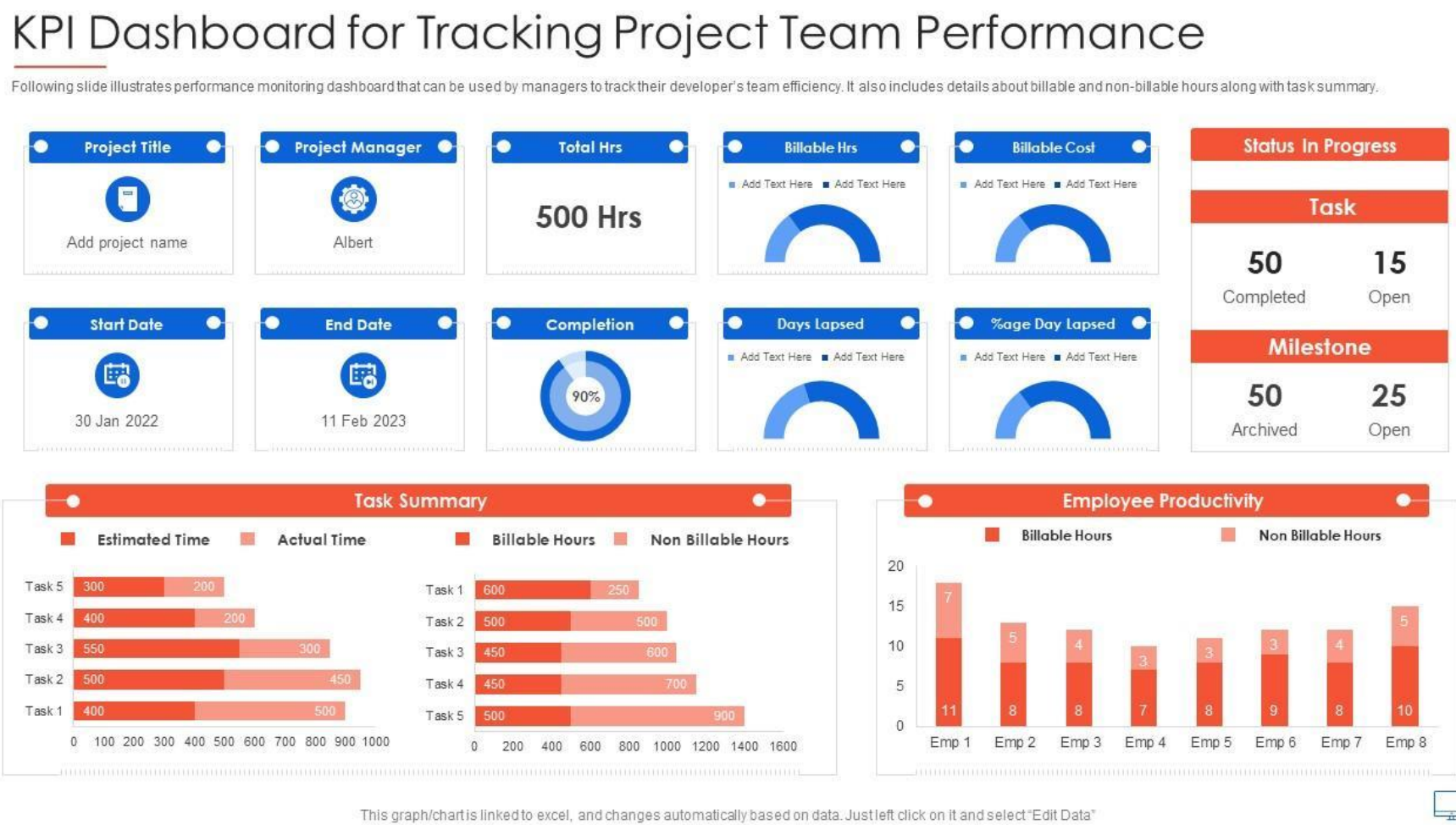


Tech Support dashboard

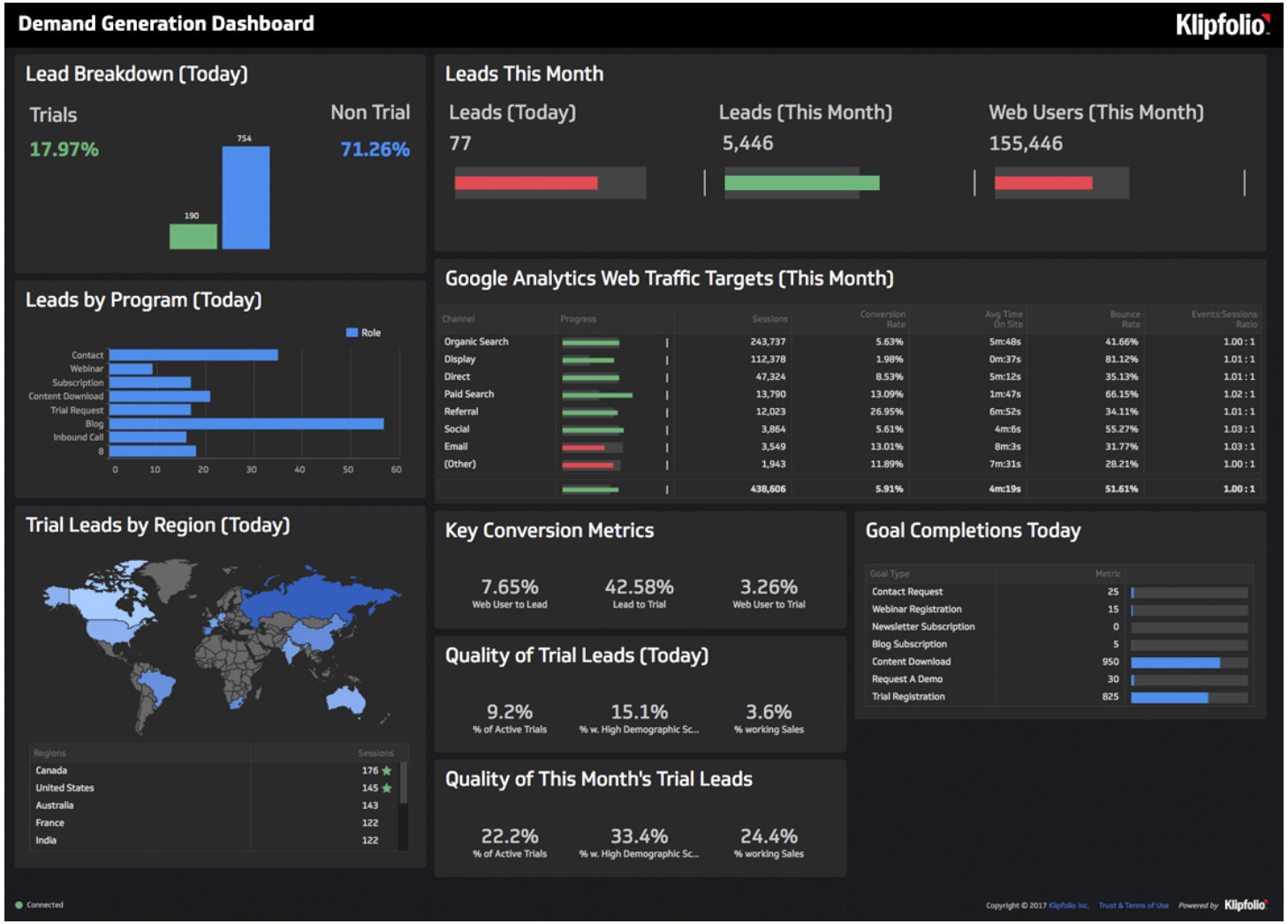
Sales Rep Dashboard



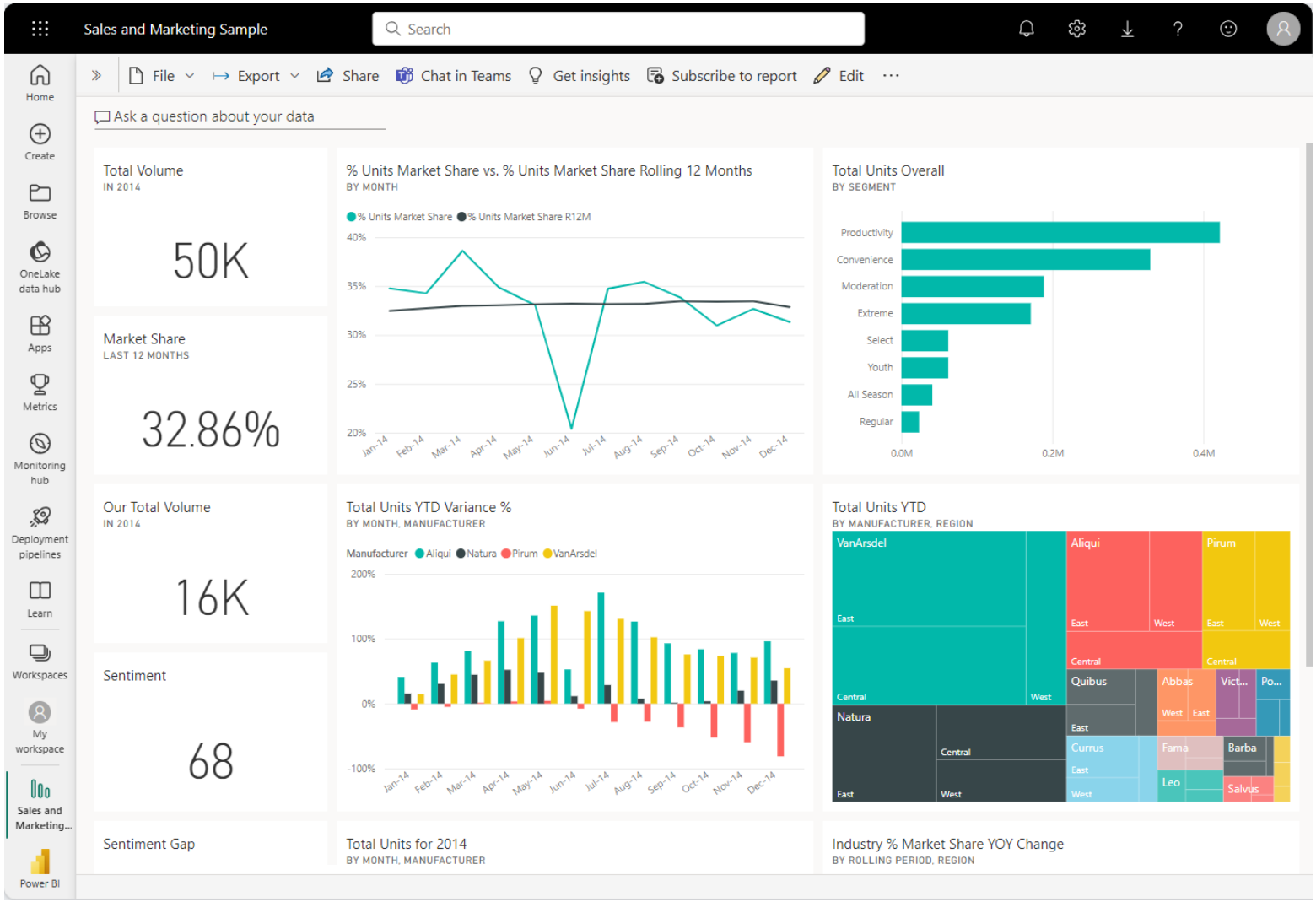
KPI Dashboard



Demand / Lead Analysis Dashboard



Sales & Marketing Dashboard



Example of a Bad Dashboard

Summary

- Studied the concept of Metrics and KPIs for defining the performance of a an aspect of the business.
- Identified various KPIs as per the business / functional activity
- Studied the working of a few KPIs
- Understood how the KPIs are represented and displayed in card or gauge vislualizations

Questions?